

Name:

Form:

MYP 2 Year 8

Plants and Ecology

Statement of Inquiry:	Exploring relationships in the environment that result from interactions between organisms.
Key Concept:	Relationships
Related Concepts:	Environment and Interaction



4-5

8B MYP Command Terms

Term	Definition
Analyse	Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.
Apply	Use knowledge and understanding in response to a given situation or real circumstances
Describe	Give a detailed account or picture of a situation, event, pattern or process
Design	Produce a plan, simulation or model
Discuss	Offer a considered and balanced review that includes a range of arguments, factors or hypotheses. Opinions or conclusions should be presented clearly and supported by appropriate evidence
Document	Credit sources of information used by referencing (or citing), following one recognized referencing system. References should be included in the text and also at the end of the piece of work in a reference list or bibliography
Evaluate	Make an appraisal by weighing up the strengths and limitations
Explain	Give a detailed account
Formulate	Express precisely and systematically the relevant concept(s) or argument(s)
Interpret	Use knowledge and understanding to recognize trends and draw conclusions from given information
Outline	Give a brief account
Present	Offer for display, observation, examination or consideration
Recall	Remember or recognize from prior learning experiences
Select	Choose from a list or group
Solve	Obtain the answer(s) using appropriate methods
State	Give a specific name, value or other brief answer without explanation or calculation
Suggest	Propose a solution, hypothesis or other possible answer
Summarize	Abstract a general theme or major point(s)

Glossary: Plants and Ecology

WORD	MEANING
Anther	Part of the stamen. It produces pollen grains.
Carpel	Female reproductive organ found in flowers. It is made of a stigma, style and ovary.
Egg cell	The female sex cell in plants.
Embryo	Tiny plant, found inside a seed, with a very small shoot and a very small root.
Fertilization	Joining of a male sex cell with a female sex cell.
Filament	Part of the stamen. It supports the anther.
Flower	Organ system containing reproductive organs – carpel (female), stamen (male).
Fruit	Something used to carry seeds. Can be fleshy or dry.
Ovary	Part of the carpel. It contains ovules, each of which contains an egg cell.
Ovule	Contains egg cells. Is found in the ovary.
Petal	Part of the flower that is usually colourful to attract pollinators.
Pollen grain	The male sex cell in plants.
Pollen tube	Tube that grows from a pollen grain down through the stigma and style and into the ovary.
Pollination	Transfer of pollen from an anther to a stigma.
Seed	Contains a plant embryo and a store of food.
Sepal	Usually green, it protects the flower bud and supports the petals.
Stamen	Male reproductive organ found in flowers. It is made of an anther and a filament.
Stem	A plant organ used to take water to the leaves and to support the leaves.
Stigma	Part of the carpel. It is where pollen lands.
Style	Part of the carpel connecting the stigma to the ovary.
Fertilizer	A chemical or natural substance that contains nutrients for plant growth.
Germination	The process by which a plant grows out from a seed.
Seed	Contains a plant embryo and a store of food.
Seed coat	Hard outer covering of a seed
Cell wall	Tough wall around plant cells. Help to support the cell.
Chlorophyll	Green substance found inside chloroplasts. It absorbs light.
Chloroplasts	Green disc containing chlorophyll. Found in plant cells. Where glucose is made by photosynthesis.

Fats	Substance needed by living things to make cell membranes. Our bodies also use fats as a store of energy and to keep warm.
Glucose	A type of sugar made during photosynthesis.
Iodine	A yellow brown solution that turns blue-black in the presence of starch.
Photosynthesis	Process that plants use to make their own food. It needs light to work. Carbon dioxide and water are used up. Glucose and oxygen are produced.
Proteins	A substance made from amino acids. Proteins are needed for growth and repair.
Starch	Insoluble carbohydrate made from glucose and used as a storage material in plants.
Accuracy	How close the measured value is to the known value.
Control variables	The variables or factors that are kept the same in a scientific investigation for fair testing.
Dependent variable	The variable that is measured in a scientific investigation.
Elodea	A type of pondweed that lives in fresh water.
Fair test	A controlled investigation carried out to answer a scientific question.
Independent variable	The variable that is changed in a scientific investigation in order to look at its effect.
Rate	How fast something is changing, measured over a period of time.
Reliability	How much you can trust the data collected.
Biofuel	A fuel that is made from plants or animal biomass.
Carbon dioxide	A gas that is a reactant in photosynthesis and a product of respiration.
Global warming	A long term rise in the Earth's temperature.
Greenhouse effect	Trapping of sun's heat on the Earth's surface.
Palm oil	A vegetable oil that comes from the fruit of oil palm trees, used widely in production of household products.
amphibian	Vertebrate with moist skin, e.g. frog.
animal kingdom	The group of organisms that contains all vertebrates and invertebrates.
arthropod	Invertebrate that has jointed legs; e.g. fly, spider.
bird	Vertebrate with feathers, e.g. eagle.
classification	Sorting things into groups.
cone	Something used to carry the seeds of conifers.
conifer	Plant with needle-shaped leaves. Reproduces using seeds found in cones.
cuticle	Layer of cells on leaves that is waterproof.
echinoderm	Invertebrate that has a body in five parts, e.g. starfish.
exoskeleton	Thick outer covering found in arthropods.
fern	Plant that has many small waterproof leaves. Reproduces using spores.
fish	Vertebrate with wet scales, e.g. salmon.

flowering plant	Plant with large, flat leaves. Reproduces using seeds found in fruits. Fruits and seeds form inside flowers.
fruit	Something used to carry the seeds of flowering plants. Can be fleshy or dry.
habitat	The place an organism lives in, e.g. woodland.
invertebrate	Animal without a backbone.
mammal	Vertebrate with hair and produces milk, e.g. human.
mollusc	Invertebrate that crawls on a fleshy pad, e.g. snail.
moss	Plant with many thin leaves but without roots and xylem. Reproduces using spores.
plant kingdom	Group of organisms that are able to produce their own food and have specialised cells.
reptile	Vertebrate with dry scales, e.g. snake.
root	Plant organ used to take water out of the soil.
seeds	Grow into new plants. Made by conifers and flowering plants.
spore	Very small part of a plant that can grow into a new plant. Made by mosses and ferns.
vertebrate	Animal with a backbone.
xylem vessel	Transports water through a plant.
adapted	When the features of an organism help it to survive in a habitat, it is adapted to that habitat.
community	All the plants and animals that live in a habitat.
environment	The conditions around an organism.
humidity	The amount of water vapour in the air.
physical environmental factors	The non-living conditions in the environment of an organism, e.g. temperature, light.
population	Total number of individuals of the same species in a habitat.
adaptation	The features of an organism that allow it to live in its environment.
compete	All organisms need some of the same things and so each organism has to try to get these things before another organism does. For example, plants compete with one another for light.
competition	Organisms compete with each other for food, light and space in a habitat.
distribution	The places where an organism can be found in a habitat.
ecologist	A person who studies the environment.
estimate	Provide a rough idea about the numbers of something or the size of something.
pitfall trap	Sampling method used to collect small animals that live on the ground.
pond dipping	Sampling method used to collect organisms from ponds.
quadrat	A square frame, thrown randomly on the ground, which is used to sample the plants in an area.

sample	A small part of something. If you sample something you take a small part of it. You use your results from the small part to suggest what the rest of it is like.
sweepnet	Sampling method used to collect small animals from long grass.
tree beating	Sampling method used to collect animals from trees and bushes.
Tullgren funnel	Sampling method used to collect small animals from samples of, for example, leaves.
uneven distribution	When a plant or animal is not found all over a habitat, only in certain places where the habitat is suitable.
pooter	A small container connected to two tubes. Used to catch tiny animals.
chemical energy	The kind of energy stored in chemicals. Food, fuels and electrical cells all contain chemical energy.
consumer	An organism that has to eat other organisms to stay alive. Animals are consumers.
food web	Many food chains linked together.
primary consumer	The first animal in a food chain.
producer	An organism that is able to make its own food.
pyramid of numbers	Way of showing the numbers of different organisms in a food chain.
secondary consumer	The second animal in a food chain.
tertiary consumer	The third animal in a food chain.
top predator	The last animal in a food chain.

8B 1a Flower Structure

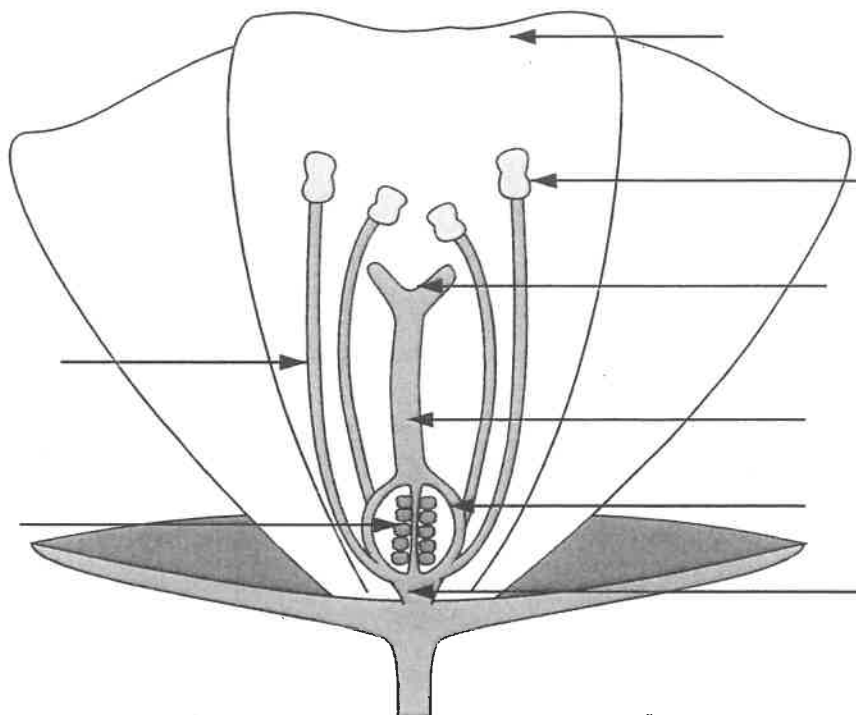
Label the diagram using the keywords from the list of structures below

Diagram labels (keywords 1-9):

1 filament, 2 anther, 3 ovary, 4

stigma, 5 petal, 6 stamen, 7

carpel, 8 stem, 9 sepal.



6 - Stamen (made up of anther and filament)

9 - Sepal (special leaves forming a ring at the bottom; protect the bud)

8 - Stem (top of the stalk where the flower parts are attached)

Write the matching number in the function column which corresponds to the structure number

Number	Structure
1	Filament
2	Anther
3	Ovary
4	Stigma
5	Petal
6	Stamen
7	Carpel
8	Stem
9	Sepal
10	Style
11	Receptacle
12	Ovule

Function	Number
Green stalk. Provides support.	
Female part, seeds made here. Consists of stigma, style, ovary.	
Supports & holds anther. Part of stamen.	
Male part of flower. Makes pollen. Made up of anther & filament.	
Special leaves form ring around bottom. Protect the buds.	
Sticky to catch pollen	
Top part of stalks where flowers attached.	
Part of stamen, contains pollen.	
Brightly coloured	
Eggs stored here, join with pollen.	
Contains female reproductive cells	
Supports stigma	

8 - Stem (green stalk that supports the plant)

3 - Ovary (female part; the ovules/seeds are made here)

7 - Carpel (female part, consists of stigma, style and ovary)

1 - Filament (supports and holds the anther; part of the stamen)

6 - Stamen (male part of the flower; makes pollen, made of anther and filament)

8B 1b Flower Dissection

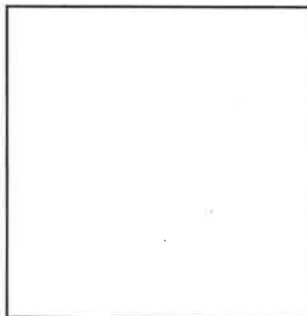
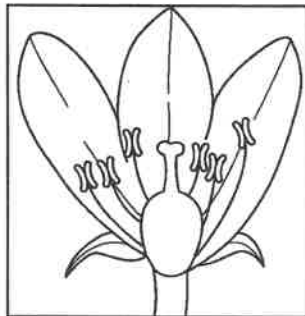
Flowers contain the male and female reproductive organs of plants.

- 1 Look carefully at your flower. What plant is it from? *Name of flower you were given, e.g. lily, daffodil or tulip.*

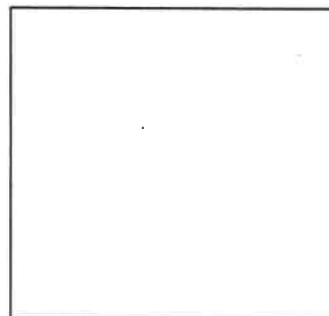
- 2 How many petals are there on your flower? *Count the petals on your own flower, e.g. 3, 5 or 6.*

- 3 What colour are the petals? *The colour you observed, e.g. yellow, white or pink.*

Using tweezers, carefully take off the petals from one side of the flower so that you can see the male and female reproductive organs. Your flower may look similar to the one below.

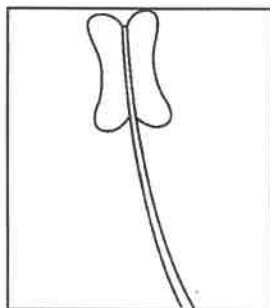


A petal from the flower – stick in or draw the shape of one petal in this box.

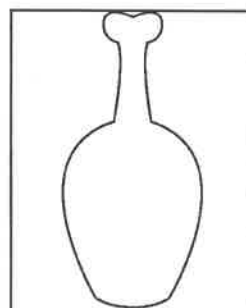


The flower looks like this – draw a picture showing the flower with half the petals taken off.

Now take off one stamen and remove the carpel. Stick them or draw them in the boxes. Label their parts if you can.



The stamen looks like this.



The carpel looks like this.

• Stick or draw one petal in the box and label it 'petal'; in the other box draw the flower with half the petals removed.

• Stamen box: anther (top, covered in pollen) and filament (stalk). Carpel box: stigma (sticky top), style (stalk) and ovary (swollen base).

4 How many stamens are there in the flower?

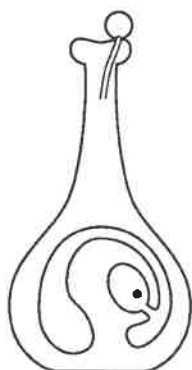
Count the stamens on your flower, e.g.
6 (look for the anthers on their stalks).

5 Any other interesting observations?

e.g. the anthers are covered in yellow pollen dust and the stigma feels sticky.

8B 1c Plant Fertilisation

- 1 The drawing below shows a pollen tube starting to grow.



It then grows towards an ovule. Here it finds an egg cell.
The nucleus from the pollen then joins with the nucleus in the egg cell.
This is called fertilisation. An embryo now grows inside the ovule.
The ovule turns into a seed.

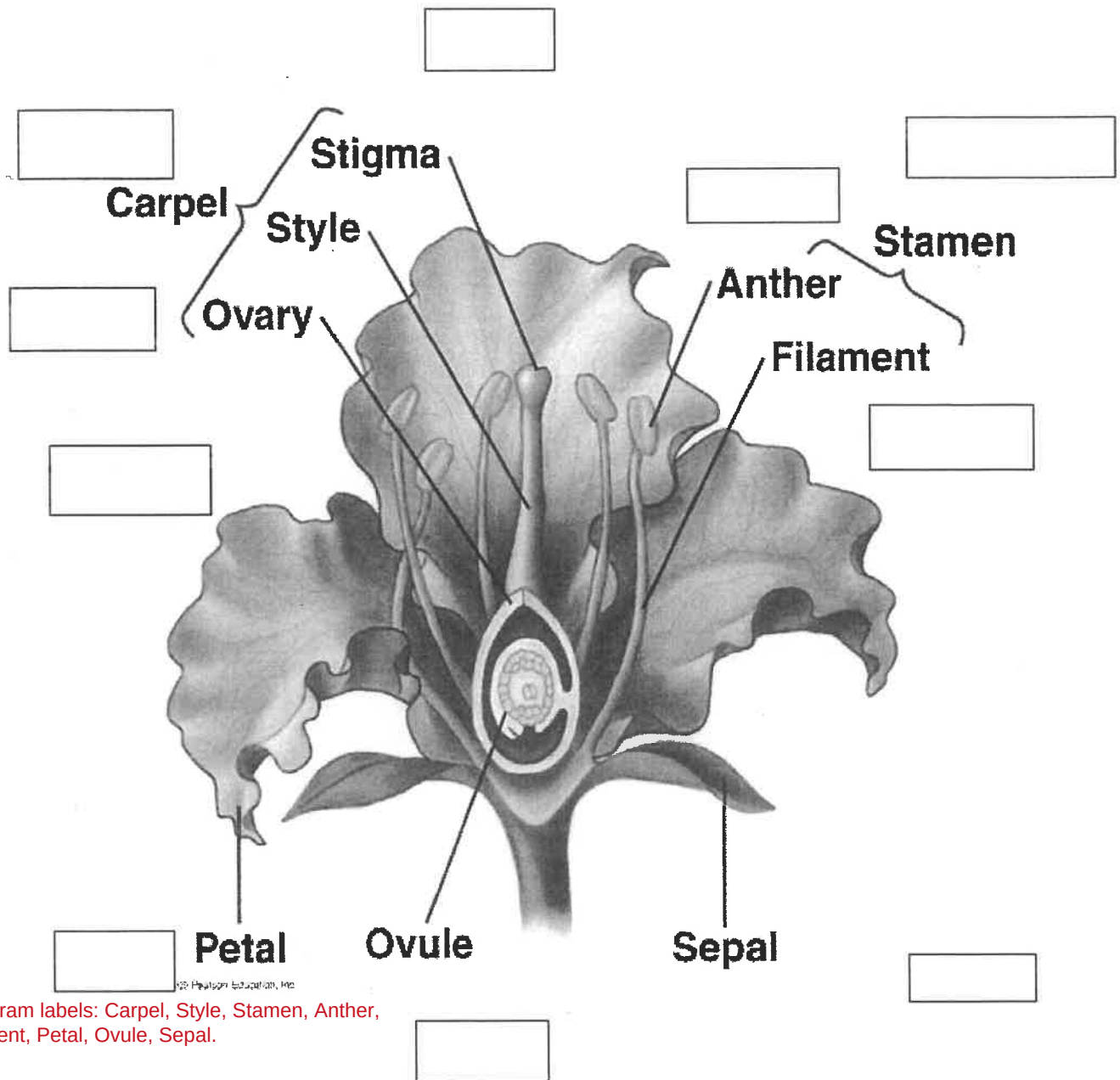
- a Complete the drawing to show where the pollen tube will end up.
b Label the parts of the diagram. Use the words in the box.

egg cell ovary ovule pollen grain pollen tube stigma style

- 2 Fill in the missing words in the following sentences.
Use the words in the box. Each word may be used more than once.

egg embryo fertilisation nucleus ovule pollen seed stigma style tube cell

When a _____ grain lands on a _____, a pollen tube starts to grow.
This grows down the _____ and into the ovary. It then grows
towards an _____. Here it finds an _____ cell. The nucleus from
the pollen _____ then joins with the _____ in the
_____ cell. This is called _____. An _____ now grows
inside the ovule.
The ovule turns into a _____.



8B 2a Factors Affecting Seed Germination

Investigating factors that affect the rate of seed Germination

Aim: To design and carry out an experiment to investigate a factor which could affect the number and/or height of seeds that germinate.

Prior Knowledge – what do you know?

Materials required

- cotton wool
- petri dish
- cress seeds
- lamp
- different NPK fertilisers
- thermometer

Seeds need water, oxygen and a suitable warm temperature to germinate; seedlings then need light to photosynthesise and grow.



Safety.

Wear safety goggles.

Brainstorm

- What factors **could** I change to increase the number/height of germinated seeds?

- Which one **will** I choose to change?

Factors: amount of light, amount of water, temperature, type of soil or fertiliser, depth of planting, number of seeds.

- How could I **measure height**?

Measure the height of each seedling with a ruler in mm, from the cotton wool to the tip of the shoot.

- What **measurements** will I make so that I have **meaningful results**?

Count how many seeds germinate each day and measure the height (from) of seeds (to) seedling after 7 days. The amount of light, e.g. 0 h, 6 h and 12 h of light a day (dark cupboard, windowsill, under a lamp).

- How will I make my experiment a **fair test**?

- How much of each substance will I use?

Keep all other variables constant: same volume of water (5 cm³), same temperature and identical dishes.

Design - Reporting

Your report should include the following:

- **Research Question** – clearly state the question you are going to investigate.
- **Explain** what you think will happen in terms of the Science involved.
- **Independent (Input) Variable** – State clearly the variable are you going to change and how will you change its value?
- **Control Variables** – state clearly the variables that you need to keep constant and say how you will control them.
- **Dependent (Outcome) variable** – state clearly what measurements you are going to take.
- **Apparatus** – draw a neat, labelled diagram of the assembled apparatus
- **Method** – write a procedure that a fellow student could follow.

Now you are ready to carry out your experiment

10 cress seeds and 5 cm³ of water in each petri dish, on damp cotton wool.

The independent variable - e.g. light intensity: move the lamp 10, 20, 30, 40 cm away from the plant, measuring each distance with a ruler.

Data Collection & Processing

- **Carry out** your planned experiment.
- **Record** your results in a *suitable table* – remember headings and units. Values of *controlled variables* and any *observations* should also be recorded.
- **Present** your results in a suitable way – line graph? Bar graph?

Concluding & Evaluating

- Use your results to draw a **conclusion**. What pattern do the results show?

Line graph if the independent variable is continuous (e.g. light intensity, distance of lamp); bar chart if the values are categories (e.g. light vs dark).

- Explain how your results helped you to reach this conclusion
- Try to **explain** the pattern using **scientific ideas**
- Describe some problems you had trying to obtain **accurate** and **reliable** results in this experiment
- Try and think of some ways of solving these problems if you were to repeat the experiment again



MYP Science Practical Report Template

Name:

Form:

UNIT:	
Statement of Inquiry:	
Key Concept:	
Related Concepts:	
Investigation Title:	Seed Germination

Experiment Design

Research Question

(What are you trying to find out?)

How does How does the amount of light affect the number of cress seeds that germinate and the height of the seedlings?

Background Scientific Information and Hypothesis

(What do you think will happen and why?)

Hypothesis: seeds given more light
will germinate in greater numbers
and grow taller because light is
needed for photosynthesis to make
glucose for growth.

Variables

Independent <i>(What will change and how will you change it?)</i> Independent variable: amount of light. Change it by placing the dishes in a dark cupboard, on a windowsill and under a lamp (12 h of light a day).	
Dependent <i>(What will you measure to gather results?)</i>	Dependent variable: number of seeds germinated (count) and height of each seedling in mm, measured with a ruler after 7 days.
Control <i>(What variables will you keep the same to make it a fair test? How will you keep the variables constant?)</i> Control variables: same type and number of seeds (10 cress), same volume of water (5 cm ³), same temperature (20 °C) and identical dishes. Use a measuring cylinder, thermometer and stopwatch to keep them constant.	

Apparatus Set-up

(Draw a labelled diagram of the assembled apparatus that you will use. Write the name of any equipment not shown in the diagram)

· Labelled diagram: petri dish, damp cotton wool, 10 cress seeds, light source (lamp/cupboard/windowsill), ruler, thermometer.

Method

(Write a step by step account of how you conduct your experiment – the method must be DETAILED). Add as many steps as you need to.

1.

1. Put the same amount of cotton wool in each of 3 petri dishes and add 5 cm³ of water with a measuring cylinder.

2.

3.

2. Put 10 cress seeds on the cotton wool in each dish, spread out evenly with tweezers.

4.

3. Put dish A in a dark cupboard, dish B on a sunny windowsill and dish C under a lamp (12 h light a day).

5.

4. Add 2 cm³ of water to each dish every day so the cotton wool stays damp.

5. After 7 days count the seeds that germinated and measure each seedling's height in mm with a ruler. Repeat 3 times.

Safety

(How will you make sure your method is safe?)

Safety: wear safety goggles, mop up spilled water so nobody slips, and take care with the lamp because it may get hot.

Reliability

(What will you do to ensure your results are reliable?)

Reliability: repeat the experiment 3 times and calculate an average; use the same number of seeds and measure in the same way each time.

Criterion C - Processing and Evaluating**Results****Raw Data**

(A neat, organized table of results should be drawn, with headings, units, repeats and averages include)

· Results table headings: Amount of light, Dish, Number germinated (repeat 1/2/3), Average number germinated, Average height after 7 days (mm).

Observations

(What changes did you notice as the experiment was conducted, e.g. colour change, bubbles, heat change?)

Observations: the seedlings in the light turned green as they grew and the cotton wool had to be watered daily; the seeds in the cupboard grew tall, thin and yellow.

Graphing Results

You should draw a graph of your results and attach it to this report.

Analysis of Graph

(What did your results show? Are there any trends or patterns in your results? Can you give an example of a pattern or trend from your results? Are there any anomalies in your results?)

· Draw a bar chart of average number germinated and average height against amount of light, with a title, labelled axes with units and a key. Attach it to the report.

Trend: all dishes germinated a similar number of seeds, but seedlings in the light grew taller and greener than those in the dark, which were yellow and weak. No obvious anomalies.

Conclusion and Evaluation

Scientific Conclusion and Explanation of Results

(Summarize what your results show. How well does this support your hypothesis ? Can you explain your results using scientific knowledge?)

Conclusion: light affects seedling growth because it is needed for photosynthesis, which makes the glucose a plant uses to grow, so seedlings kept in the dark were yellow and weak.

Evaluation of Scientific Method and Results

(How reliable were your results? How well did your investigation answer the research question? What went well with your method?)

Results were fairly reliable: each condition was repeated 3 times and averaged. The method answered the research question because only the amount of light was changed.

(Were there any problems or parts of your investigation that were not very accurate or could have been done better?)

Problems: the lamp dish dried out, seeds clumped together so they had different amounts of water, and measuring curved shoots with a ruler was inaccurate.

Issue with justification	Suggested SCIENTIFIC Improvement
Issue: the lamp dish dried out so its seeds had less water (unfair test). Improvement: add exactly 2 cm ³ of water with a measuring cylinder to every dish each day and keep all dishes at the same room temperature.	

8B 3a GM Crops - The Good, the Bad and the Ugly

Video Title	Impressions	Arguments <u>FOR</u> use of GMO	Arguments <u>AGAINST</u> use of GMO
GMO and the Environment https://www.youtube.com/watch?v=KonUXpTv0SI	<u>Credible?</u> ☒ Yes ☐ No <u>Bias</u> ☐ Pro ☐ Against ☒ Neutral	GMO and the Environment. FOR: GM crops can be made to resist pests, disease and drought, giving bigger yields and less pesticide. AGAINST: GM pollen may spread to wild plants and reduce biodiversity.	
Greenpeace https://www.youtube.com/watch?v=1H9WZGKQeYg	<u>Credible?</u> ☒ Yes ☐ No <u>Bias</u> ☐ Pro ☐ Against ☐ Neutral	Greenpeace. AGAINST: GM crops are not proven safe and could harm wildlife; there is little long-term evidence. FOR: GM crops could reduce the amount of chemical spraying.	

Bayer https://www.youtube.com/watch?v=frH8a0BYWgs	<u>Credible?</u> Yes No <u>Bias</u> ☐ Pro ☐ Against ☐ Neutral		
Fuse School https://www.youtube.com/watch?v=kZIYkYNpnP0	<u>Credible?</u> Yes No <u>Bias</u> ☐ Pro ☐ Against ☒ Neutral		Bayer (biotechnology company). FOR: GM crops are thoroughly tested and help farmers produce more food. AGAINST: a company that sells GM seeds may exaggerate the benefits. Fuse School. FOR: explains how GM crops are produced and their advantages. AGAINST: only briefly mentions the risks of genes spreading to wild plants.
Ted Ed https://www.youtube.com/watch?v=7TmcXYp8xu4&t=61s	<u>Credible?</u> Yes No <u>Bias</u> ☐ Pro ☐ Against ☒ Neutral		

Ted Ed. FOR: GM crops can increase food security and reduce hunger. AGAINST: ethical concerns and environmental risks, and seeds are patented so farmers must buy them each year.

Claim, Evidence, Reasoning

Using the information you gathered from the videos, make a personal claim, based on the role you have assumed within your group about whether or not GMO foods should be allowed in our food supply.

Provide **four** pieces of evidence and explain why/how they support your claim.

Claim: GM foods should be allowed in our food supply, but only if they are safety-tested and labelled, because they can help feed a growing world population.

Evidence 1: GM crops engineered to resist pests give higher yields, so less land is needed to grow the same amount of food.

Evidence 2: crops that need fewer pesticides mean less chemical pollution of soil, rivers and drinking water.

Evidence 3: GM foods such as golden rice contain extra vitamins, which can reduce deficiency diseases in poorer countries.

Evidence 4: there is little long-term evidence about the effects of GM food on human health or on wild plants, so caution and labelling are needed.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



MYP 2: Year 8 Science

Criterion D Assessment - Reflecting the Impacts of Science

UNIT:	Plants & Ecology
ASSESSMENT TITLE:	GM Crops: the Good, the Bad, the Ugly
Inquiry Statement	Exploring relationships in the environment that result from interactions between organisms.
Key Concept:	Relationships
Related Concepts:	Environment, Interaction

Assignment Instructions (GRASPS)

Goal	Produce a group podcast which outlines a variety of viewpoints associated with using GM crops.
Role	You must work on this task in groups of FOUR . Each person in the group will assume a different role from the following list; 1. Farmer 2. Environmentalist 3. Doctor 4. Director of a Biotechnology Firm producing GM Crops.
Audience	The local community.
Situation	The local community has seen an increased interest in the use of genetically modified crops as a food source and would like further information on the pros and cons of such crops before approval for use can be given. Your podcast will be broadcast on the local radio station.
Product	Produce a group podcast in which you discuss GM crops from the viewpoint of the role you have been assigned. The podcast should be no more than a TOTAL of 5 minutes in length, allowing each member of your group to provide a 1min 20s viewpoint.
Success Criteria for Assessment	In your podcast you will: - Clearly explain the advantages and disadvantages associated with GM crops - You refer to data to support your argument in the role you have assumed. - You must use accurate scientific terminology in your verbal communication in the podcast. - Draw a final conclusion to state that from the role you have assumed are you for or against GM crops

Other Information:

You **MUST** provide a Bibliography and cite all your sources.

Due Date: this will be set by your teacher.

Level Descriptors and Task Specific Clarifications for Criterion D:

Level	Level Descriptor	Task Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	No data or clear evidence provided.
1-2	The student is able to, with limited success : i. state the ways in which science is used to address a specific problem or issue ii. state the implications of using science to solve a specific problem or issue, interacting with a factor iii. apply scientific language to communicate understanding iv. document sources.	ii. Student simply states the problem of using non GM crops. ii. Simply state that GM crops are better. No explanation or relevant data given. iii. Little evidence of scientific vocabulary being used. iv. Source of information given, but not correctly referenced.
3-4	The student is able to: i. Outline the ways in which science is used to address a specific problem or issue ii. Outline the implications of using science to solve a specific problem or issue, interacting with a factor iii. sometimes apply scientific language to communicate understanding iv. sometimes document sources correctly.	i. At least one problem associated with GM crops. Some data is given, but source may not be referenced. ii. Students provide at least 1 reason to explain why GM crops are an improvement. iii. Some scientific vocabulary is used. iv. Bibliography has only one source referenced, but this may not be accurately formatted.
5-6	The student is able to: i. Summarize the ways in which science is used to address a specific problem or issue ii. Describe the implications of using science to solve a specific problem or issue, interacting with a factor iii. usually apply scientific language to communicate understanding clearly and precisely iv. usually document sources correctly.	i. Problems associated with non GM crops are summarized, but not explained fully. Some data/example that supports the argument is given. ii. The advantages of GM crops are described, but not fully explained. iii. Some good scientific vocabulary is used. iv. At least 3 sources are referenced in APA format
7-8	The student is able to: i. Describe the ways in which science is applied and used to address a specific problem or issue ii. Describe and summarize the implications of using science and its application to solve a specific problem or issue, interacting with a factor iii. consistently apply scientific language to communicate understanding clearly and precisely iv. document sources completely	i. The student presents a clear explanation of the problems associated with non GM crops, using examples. ii. Contrast these with the advantages of GM crops, describing and summarising how these work. Students give clear and specific examples. iii. Explanations involve excellent scientific vocabulary. iv. Correct bibliography is provided citing at least 5 reputable and relevant sources, in APA format.

8B 5a Photosynthesis – Key Information

Highlight any information that you think is relevant to how plants get the materials to grow in one colour and what happens to the materials they produce in another colour.

· Highlight in one colour how plants get materials to grow (carbon dioxide diffuses from the air through the stomata; water is taken from the soil by the roots) and in another colour what happens to the products (glucose is used in respiration or changed into starch, cellulose, proteins, oils and fats to make new cells and increase biomass).

Green plants are unique in their ability to make their own food. Unlike most organisms that have to consume food that is already there, green plants have the ability to produce their own. This is called Photosynthesis.

This is done using carbon dioxide from the air, and water from the soil. Water is transported to the leaves, where photosynthesis occurs, whilst carbon dioxide diffuses into the leaf from the air through tiny holes in the underside of the leaf called stomata.

The process then requires energy to break the atoms in carbon dioxide and water apart. This energy comes from sunlight, which plants collect. In order to harness this energy, the cells of the leaf contain chloroplasts, which have a green chemical called chlorophyll. This is the chemical that harnesses the light energy.

Once the atoms of carbon dioxide and water have been broken apart (carbon, hydrogen and oxygen), they are then rearranged to form two products, one of which is glucose. The plant can then convert glucose into many different things, including starch for storage, cellulose for new cell walls, proteins, oils and fats. All of these materials can then be used to make new cells, which in turn mean the plant grows bigger. We call the amount of biological material that the plant has, its biomass. Therefore the plant can use the food it makes to increase its biomass.

This is good news for us, as it means that animals and humans have food to eat.

8B 5b Testing for Starch in Plants

Apparatus

- Food samples
- Water
- Spotting tile
- Pipette
- Iodine solution



Iodine will stain your skin.

Method

- 1 You will be provided with a range of human foods that come from plants. For each food, write down the part of the plant that it has come from in the results table below.
- 2 Use a pipette to add two drops of iodine solution to each food.
- 3 Look for a blue-black colour. If it appears after testing, then starch is present.
- 4 Enter your results in the table below.

Recording your results

Name of food	Part of plant	Does it contain starch?

Considering your results/conclusions

Do all the parts of the plants that you tested contain starch? _____

No - only some parts contain starch. The potato and bread turned blue-black, but the apple and carrot did not.

Potato - stem (tuber) - starch: yes (blue-black). Bread - wheat seed - starch: yes. Apple - fruit - starch: no. Carrot - root - starch: no.

Do any of the food samples contain more starch than others?

Starch is a store of energy for the plant. For each food tested that contained starch, suggest a reason why a store of energy might be needed in that part of a plant.

Yes - some contain much more starch than others. Potato gave the strongest blue-black colour, so it contains the most starch.

e.g. a potato stores starch in its underground stem so the plant has an energy store to make new shoots grow in spring, before the leaves can photosynthesise.

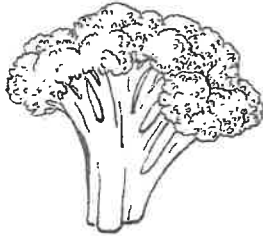
Suggest some reasons why humans might eat the foods that do not contain starch.

Foods without starch are eaten for other nutrients - vitamins, minerals, fibre, protein and water - and for their taste and texture, not for energy (e.g. fruit and vegetables).

8B 5c Plants We Eat

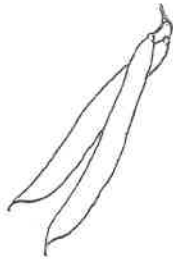
- 1 The drawings below show plants that we eat. For each one, write down the name of the plant and which part of it we eat. Use the words in the box to help you.

apple	broccoli	carrot	celery	flower	fruit	hazel nut
leaf	lettuce	potato	root	runner bean	stem	tomato



Name _____

Part of plant _____



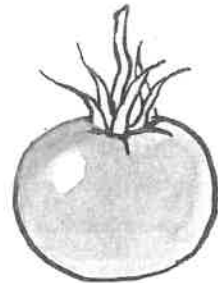
Name _____

Part of plant _____



Name _____

Part of plant _____



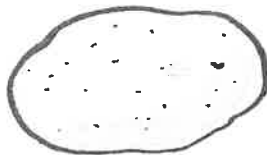
Name _____

Part of plant _____

Carrot - root; Celery - stem; Lettuce - leaf; Broccoli - flower; Apple - fruit

Potato - stem (tuber); Runner bean - fruit (pod containing seeds); Tomato - fruit; Hazel nut (seed)

Hazel nuts (or the runner bean seeds) - seeds store protein and fat for the new plant as it germinates.



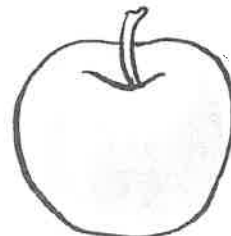
Name _____

Part of plant _____



Name _____

Part of plant _____



Name _____

Part of plant _____

- 2 All the materials in all the foods are made using glucose. What process do plants use to make glucose? photosynthesis.

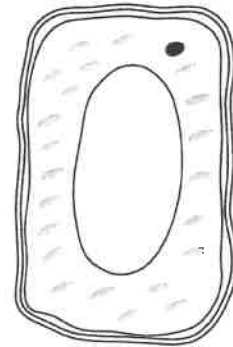
- 3 Some glucose is turned into proteins and fats. These are often stored in seeds. Choose one food from question 1 which you would expect to contain a lot of protein and fat.

- 4 Glucose is often turned into starch. Choose one food from question 1 which contains a lot of starch. Potato - it stores a lot of starch in its stem tuber.

8B 5d Uses of Glucose

As a store of energy for the plant to use later, e.g. for growth or when it cannot photosynthesise.

- 1 Copy this drawing of a cell in a leaf. It is called a palisade cell and it makes lots of glucose by photosynthesis.



- 2 What process happens in the cell to make glucose?
- 3 Most of the glucose is changed into starch as soon as it is made. Label your palisade cell drawing to show where you would expect to find starch.
- 4 Some plants can store starch in special parts. Which one of these stores a lot of starch?

Photosynthesis.

· Label the chloroplasts (the green discs) - this is where glucose is made and starch is stored.

orange apple flower potato tomato pear

- 5 People eat food with starch in it to give them energy. Why do plants store up starch?
- 6 Glucose molecules are shaped like this:



potato

Many glucose molecules join in a line to form a starch molecule. Make a drawing of what you think a starch molecule looks like. The links between each glucose molecule can be shown by a straight line.

- 7 Some of the glucose is used to make cellulose. Label your palisade cell drawing to show where you would expect to find cellulose.
- 8 Some of the glucose is used to make fats and oils. These are needed to make cell membranes. Label your palisade cell drawing to show where you would expect to find oils and fats.
- 9 Glucose is used for aerobic respiration in plants. Copy and complete the word equation for aerobic respiration.

_____ + _____ → carbon dioxide + water (+energy)

- 10 Aerobic respiration happens in the cytoplasm of every cell. Label your palisade cell drawing to show where you would expect to find respiration happening.

· Drawing: many hexagon-shaped glucose units joined in a long line like a chain, with a straight line drawn between each glucose molecule.

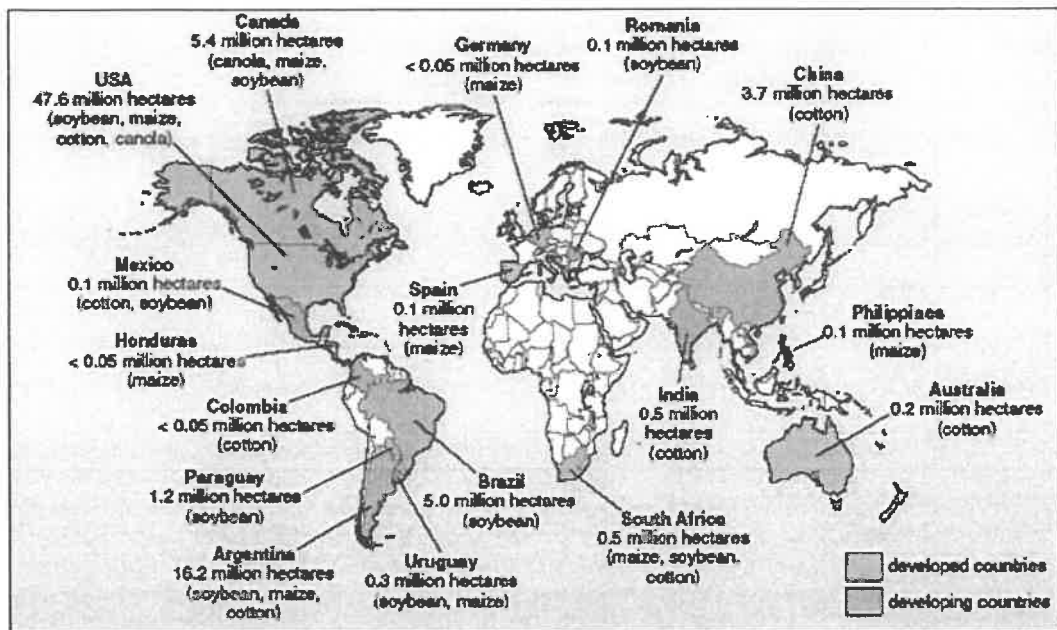
· Label the cell wall - cellulose is found in the cell wall.

· Label the cell membrane (and cytoplasm) - oils and fats are used to make cell membranes.

glucose + oxygen → carbon dioxide + water (+ energy)

· Label the cytoplasm - aerobic respiration happens in the cytoplasm of every cell.

8B 5e Crops of the World



Q1) Define a crop

Q2) Name 3 different categories of crops

Q3) Complete the following table on major crops

Name	Details on nutritional content	Countries where this is grown	Additional information
Wheat	A crop is a plant that farmers grow and harvest in large quantities, usually for food, fuel or other uses.	Food crops (cereals, vegetables, fruits), oilseed crops, fibre crops (e.g. cotton), and biofuel crops.	
Rice		Rice: carbohydrate; China, Philippines; flooded paddy fields. Maize: starch; Canada, China, Australia; food and animal feed. Soybean: protein/oil; China, Brazil. Cotton: fibre; China, Australia, Brazil. Canola: oil; Canada.	
Maize			

Q4) Complete the following table on 3 additional crops of your choice

Name	Details on nutritional content	Countries where this is grown	Additional information
	Banana: carbohydrate and potassium; Philippines, Brazil; tropical climate. Wheat: starch; China, USA; made into flour/bread. Sugarcane: sugar; Brazil, India; used to make ethanol fuel.		

Keywords (add any extra to this list as you find them)

Biodiversity, food, grazing, agriculture, arable, farming, plants, crops, harvested

Useful websites

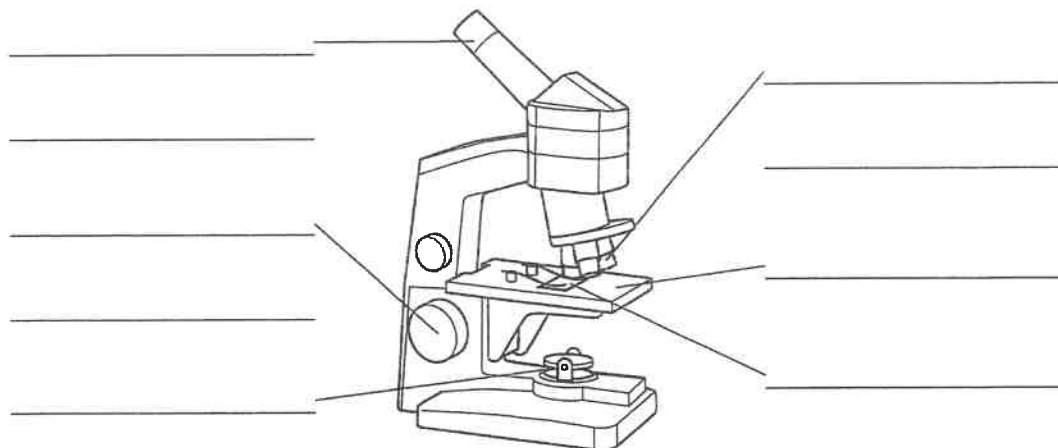
<https://www.idrc.ca/en/article/facts-figures-food-and-biodiversity>

<http://nationalgeographic.org/encyclopedia/crop/>

8B 6a Microscope Review

Label the parts of a light microscope.

· Label the parts of the microscope: eyepiece, objective lenses, stage, stage clips, coarse and fine focus knobs, arm, base and mirror/illuminator.



The eyepiece of the microscope we use has a magnification of 10x. Calculate the total magnification of images seen using different objective lenses.

Hint: Total magnification = Eyepiece magnification x Objective lens magnification

Objective lens	Objective lens magnification	Eyepiece magnification	Total magnification
Lower power			
Medium power			
Higher power			

4x objective -> 40x total; 10x objective -> 100x total; 40x objective -> 400x total.

Total magnification = eyepiece magnification (10x) x objective lens magnification.

8B 6b Starch Granules and Chloroplasts

Observing chloroplasts in pondweed

Pondweeds such as elodea are suitable to be viewed under the microscope as their leaves are thin and therefore let sufficient light through.

How can you identify the chloroplasts?

Materials:

Pondweed, glass slide, coverslip, pipette and water

Method:

1. Use scissors or your fingers to gently remove one leaf from the pondweed.
2. Place the leaf on a glass slide.
3. Add one drop of water on the leaf.
4. Place coverslip over the specimen.
5. View the leaf under lower and medium power on the microscope.
6. Identify the cell wall and chloroplasts in the pondweed cells.

In the space below, produce a magnified drawing of one pondweed cell seen under medium power. Label the parts visible.

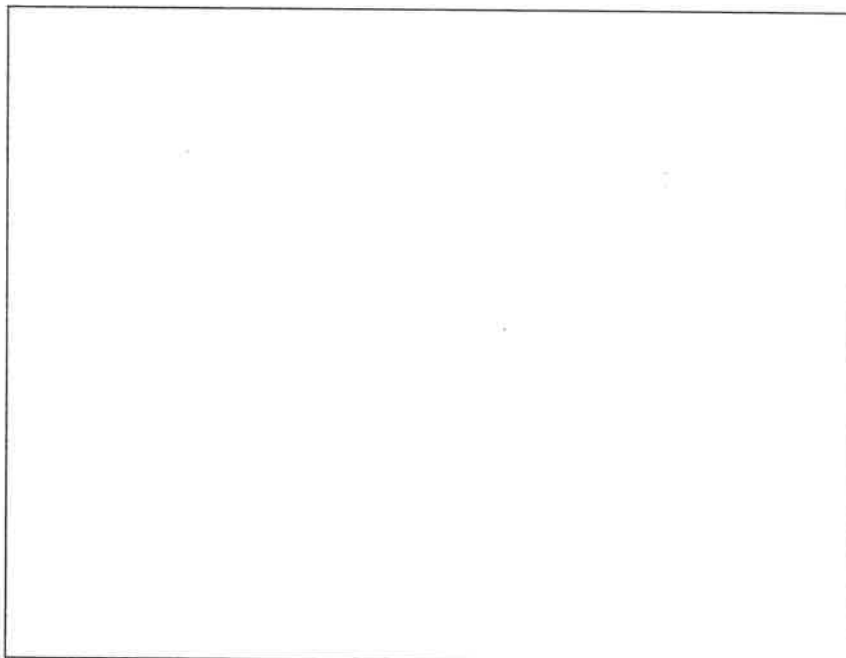
· Draw one pondweed cell seen under medium power using a pencil and solid lines; label the cell wall and the chloroplasts (green oval granules).

Tips for good biological drawings:

- Use a pencil
- Use single, solid lines
- No shading
- Draw only what you can see



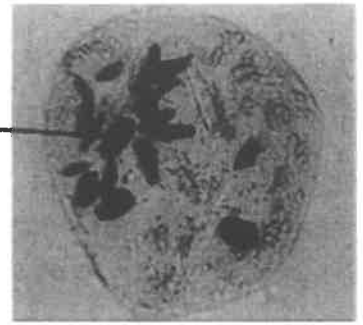
Chloroplasts look like small green oval granules inside the cells; they contain green chlorophyll, which is why they are green.



Observing starch granule in banana

Plants such as banana stores food in the form of starch in their cells.

Starch granule

***How can you identify starch granules?***

Starch granules are stained blue-black by iodine solution, so they appear as dark blue-black grains inside the cells.

Materials:

Unripe banana, glass slide, coverslip, iodine solution, toothpick

Source: <https://www2.mrc-lmb.cam.ac.uk/microscopes4schools/banana.php>

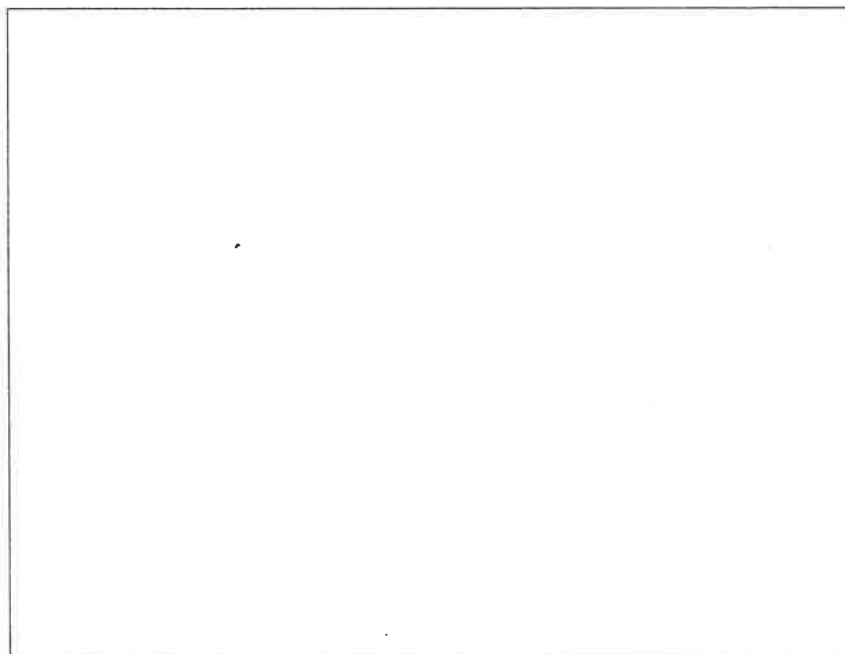
Safety goggles should be worn as iodine solution is an irritant

Method:

(adapted from <https://www2.mrc-lmb.cam.ac.uk/microscopes4schools/banana.php>)

1. Collect a small piece of unripe banana.
2. Using a toothpick, smear a little bit (less than the size of a sesame seed) of the banana on a microscope slide to rub the cells apart.
3. Add one drop of iodine solution on the specimen and place coverslip over.
4. Remove excess solution with a piece of tissue paper.
5. View the slide under lower and medium power on the microscope.
6. Identify starch granules which are stained blue-black.

In the space below, produce a magnified drawing of one banana cell seen under medium power. Label the parts visible.



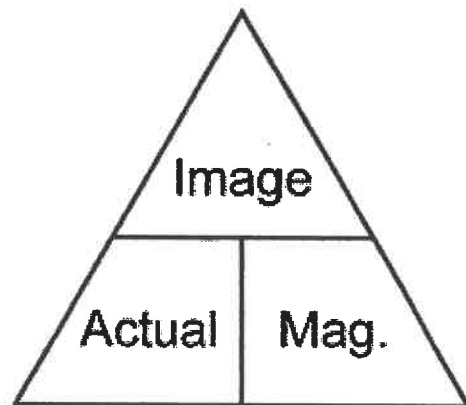
· Draw one banana cell seen under medium power; label the starch granules (stained blue-black) and the cell wall.

8B 6c Magnification Calculations (Extension)

The magnification of an image can be calculated using the formula:

$$\text{Magnification} = \frac{\text{Image size}}{\text{Actual size}}$$

$$1,000\mu\text{m} = 1\text{mm}$$



Calculate the **magnification** of the following images:

1. If the sperm cell is $53\mu\text{m}$ in length (head to tail)



Measure the drawn sperm cell image length, convert to μm ($1\text{mm} = 1000\mu\text{m}$), then magnification = image size / actual size ($53\mu\text{m}$).

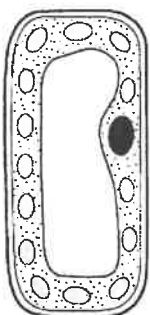
2. If the red blood cell is $8\mu\text{m}$ in diameter

Measure the drawn cell diameter in mm and convert to μm ; magnification = image size / actual size ($8\mu\text{m}$).



3. If the largest chloroplast in this palisade cell is $5\mu\text{m}$ in length.

Measure the drawn chloroplast length in mm and convert to μm ; magnification = image size / actual size ($5\mu\text{m}$).



Calculate the **actual size** of the following cells or structures:

4. If the **white blood cell** diameter is magnified $\times 2700$

Measure the drawn white blood cell diameter in mm, convert to μm ; actual size = image size / 2700.



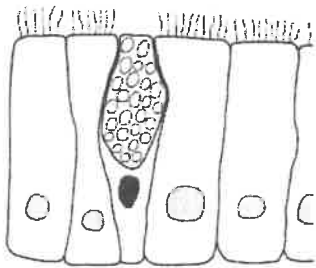
5. If the diameter of the **xylem vessel** is magnified $\times 650$



Measure the drawn xylem vessel diameter in mm, convert to μm ; actual size = image size / 650.

6. If the length of the longest **cilia** are magnified $\times 2000$

Measure the drawn cilia length in mm, convert to μm ; actual size = image size / 2000.



8B 7a Investigate a Factor that Affects the Rate of Photosynthesis Criteria B & C Assessment Overview

Let's review our Statement of Inquiry.

Exploring **relationships** in the **environment** that result from **interactions** between organisms.

Photosynthesis is fundamental to not just plants but all living things. Oxygen is released into the atmosphere through photosynthesis to be used by other organisms for respiration. Plants are a food source for many organisms. Factors that affect the rate of photosynthesis are therefore of great significance.

Aim

To investigate a factor that affects the rate of photosynthesis in *Elodea*.

Tasks

i) Planning

- You will work in groups of 2 or 3 and agree on **ONE factor** to investigate.
- Your teacher will demonstrate a method to measure the rate of photosynthesis, by counting the **number of bubbles** rising from the cut piece of *Elodea*.
- You need to **complete** the planning of your investigation before starting data collection. You may have time to do some pre-testing to finalise your method.
- You can use the planning board to help you design your investigation and identify the **independent, dependent and controlled variables**.

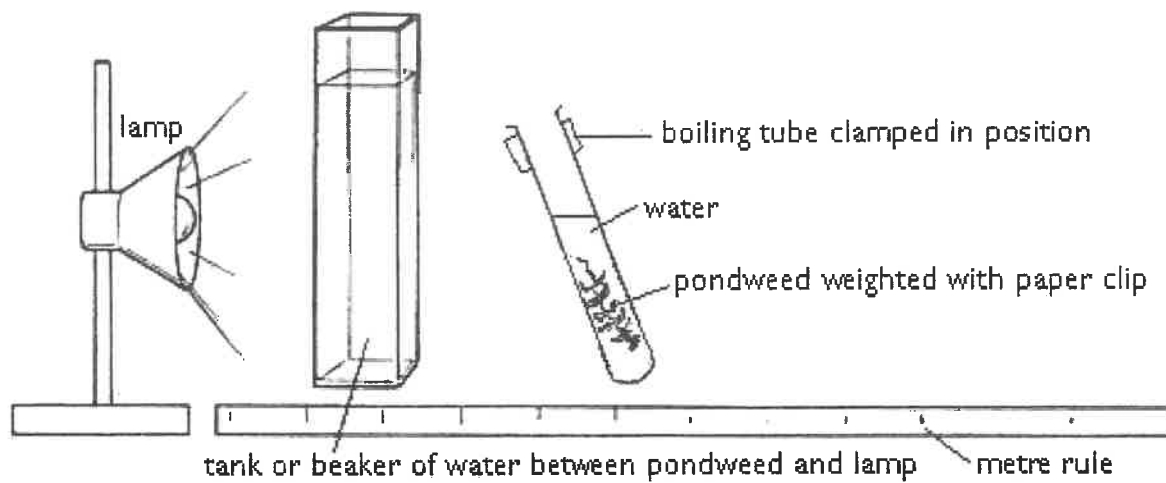
ii) Data collection

- You should try to collect as much data as you can within the time given in class. Remember **multiple trials** help increase the reliability of your data.
- Make sure a **results table** is set up for data to be recorded.

iii) Write up

- You need to produce a **full scientific report** to document your planning and findings.
- You could use the report **template** attached to help structure your work.
- Make sure you check your work against the **checklist**.

Adapted from <http://www.nuffieldfoundation.org/practical-biology/investigating-factors-affecting-rate-photosynthesis>

Set up

Sodium hydrogen carbonate (0.5g at a time) could be added to alter the carbon dioxide concentration.

Planning Notes


MYP 2: Year 8 Science
Criterion B Assessment – Inquiring and Designing
Criterion C Assessment – Processing and Evaluating

UNIT	Plants and Ecology																	
ASSESSMENT TITLE:	Investigate into a Factor Affecting the Rate of Photosynthesis																	
Statement of Inquiry:	Exploring relationships in the environment that result from interactions between organisms.																	
Key Concept:	Relationships																	
Related Concepts:	Environment, Interaction																	
ATL Focus:	Social – collaboration (ii) – manage and resolve conflict, and work collaboratively Self-management – organisation (iii) – select and use technology effectively and productively																	
Key Scientific Vocabulary:	<table><tr><td>Prediction</td><td></td></tr><tr><td>Independent variable</td><td></td></tr><tr><td>Dependent variable</td><td></td></tr><tr><td>Controlled variables</td><td></td></tr><tr><td>Reliability</td><td></td></tr><tr><td>Qualitative data</td><td></td></tr><tr><td>Quantitative data</td><td></td></tr><tr><td>Validity</td><td></td></tr></table>		Prediction		Independent variable		Dependent variable		Controlled variables		Reliability		Qualitative data		Quantitative data		Validity	
Prediction																		
Independent variable																		
Dependent variable																		
Controlled variables																		
Reliability																		
Qualitative data																		
Quantitative data																		
Validity																		
Command Terms:	<table><tr><td>Design</td><td>Produce a plan, simulation or model</td></tr><tr><td>Outline</td><td>Give a brief account</td></tr><tr><td>Interpret</td><td>Use knowledge and understanding to recognize trends and draw conclusions from given information</td></tr><tr><td>Analyse</td><td>Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.</td></tr></table>		Design	Produce a plan, simulation or model	Outline	Give a brief account	Interpret	Use knowledge and understanding to recognize trends and draw conclusions from given information	Analyse	Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.								
Design	Produce a plan, simulation or model																	
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Analyse	Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.																	

Assignment Instructions (GRASPS)

Goal	To investigate how ONE factor affects the rate of photosynthesis in elodea.
Role	Students are acting as scientists to report on findings from a scientific investigation.
Audience	The audience of this report is a group of environmental scientists who have been studying a nearby lake.
Situation	Environmental scientists are interested in finding out reasons for the recent changes in oxygen level in the lake, and have asked for your help to investigate.
Product	A scientific investigation report to communicate your findings.
Success Criteria for Assessment	The <i>final scientific report MUST include</i> (see MYP Practical Report Checklist for Students in your booklet): - Title - Research Question - Background information - Hypothesis - Variables - Apparatus - Method - Safety considerations - Results table - Observations - Graph - Analysis and Conclusion - Strengths and Weaknesses - Improvements

Level Descriptors and Task Specific Clarifications for Criterion B:

Level	Level Descriptor	Task Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	i. No research question stated. ii. No prediction stated. iii. Variables not identified. iv. Method and safety precautions not stated.
1-2	The student is able to: i. select a problem or question to be tested by a scientific investigation ii. select a testable prediction iii. state a variable iv. design a method with limited success .	i. A research question is stated but not relevant to the investigation. Limited background information on photosynthesis given. ii. A prediction with reference to photosynthesis is stated . iii. Independent; dependent and controlled variables identified but with inaccuracies. Method of data collection not indicated. iv. A method is presented but incomplete or not relevant to the investigation. V. Safety precautions not stated.
3-4	The student is able to: i. state a problem or question to be tested by a scientific investigation ii. outline a testable prediction iii. outline how to manipulate the variables, and state how data will be collected iv. design a safe method in which he or she selects materials and equipment .	i. A research question is stated with reference to photosynthesis . Background information outlines the reaction of photosynthesis . ii. A prediction with reference to photosynthesis is stated with limited explanation. iii. Independent, dependent and controlled variables identified but incomplete. Basic method of data collection is stated . iv. Method is presented in steps but does not clearly show how the experiment is conducted and/or how data is collected. Safety precautions stated but are not specific to the experiment.
Level	Level Descriptor	Task Specific Clarification

5-6	The student is able to: i. outline a problem or question to be tested by a scientific investigation ii. outline and explain a testable hypothesis using scientific reasoning iii. outline how to manipulate the variables, and outline how sufficient relevant data will be collected iv. design a complete and safe method in which he or she selects appropriate materials and equipment.	i. A research question is stated with reference to photosynthesis. Background information outlines the reaction of photosynthesis, and how the rate of photosynthesis could be compared. ii. A prediction with reference to photosynthesis is outlined and explained with basic reasoning. iii. Independent, dependent and controlled variables correctly identified, with an outline of how relevant data is collected to measure photosynthesis. iv. Method is presented in clear steps with relevant details, with suitable materials and apparatus chosen. A labelled diagram to show how the apparatus is set up. Relevant safety precautions discussed.
7-8	The student is able to: i. describe a problem or question to be tested by a scientific investigation ii. outline and explain a testable prediction using correct scientific reasoning iii. describe how to manipulate the variables, and describe how sufficient, relevant data will be collected iv. design a logical, complete and safe method in which he or she selects appropriate materials and equipment.	i. A research question clearly identifies the independent and dependent variables with reference to photosynthesis. Background information outlines the reaction of photosynthesis, factors that could affect the reaction, and how the rate of photosynthesis could be compared. ii. A prediction with reference to the relationship between the independent and dependent variables is outlined and explained using relevant scientific reasoning. iii. Independent, dependent and controlled variables correctly identified; outlining how they are manipulated, measured or controlled, and why. Values of variables stated where appropriate. iv. Method is presented in clear steps with all relevant details, stating values and measuring equipment used. A fully labelled diagram to show how the apparatus is set up. Safety precautions discussed and are relevant to this investigation.

Level Descriptors and Task Specific Clarifications for Criterion C:

Level	Level Descriptor	Task Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	i. No data presented. No graph plotted. ii. Trend not stated. iii. No conclusion on whether the prediction is supported by data. iv. No strengths or limitations relevant to the investigation stated. v. No improvement or extension stated.
1-2	The student is able to: i. collect and present data in numerical and/or visual forms ii. Accurately interpret data iii. state the validity of a prediction based on the outcome of a scientific investigation, with limited success iv. state the validity of the method with limited reference to a scientific investigation v. state limited improvements or extensions to the method	i. Raw, quantitative data from are presented in a table. A graph is plotted with inaccuracies. ii. A trend is stated but is not supported by data. No scientific explanation given to explain the trend. iii. A conclusion is drawn to state if the prediction is supported by the data, but with inaccuracies. iv. At least one limitation relating to the investigation is stated, but not relevant to the validity of the investigation. v. At least one improvement or extension stated but is not relevant to the limitation identified.
3-4	The student is able to: i. correctly collect and present data in numerical and/or visual forms ii. accurately interpret data and describe results	i. Raw, quantitative data are neatly presented in a table with headings. Averages calculated and presented in a table, with some inaccuracies. A graph is plotted with labelled axes. Limited qualitative data from the experiment recorded. ii. Trend shown in the graph is outlined with some inaccuracies. Limited explanation on the trend is given and is not fully relevant to the chosen factor and photosynthesis.

	iii. state the validity of a hypothesis based on the outcome of a scientific investigation iv. state the validity of the method based on the outcome of a scientific investigation v. state improvements or extensions to the method that would benefit the scientific investigation.	iii. A conclusion is drawn to state if the prediction is supported by the data. iv. At least one limitation relating to the validity of the investigation is stated. v. At least one improvement relating to each limitation raised is stated. Extensions of investigation stated.
5-6	The student is able to: i. correctly collect, organize and present data in numerical and/or visual forms ii. accurately interpret data and describe results using scientific reasoning iii. outline the validity of a hypothesis based on the outcome of a scientific investigation iv. outline the validity of the method based on the outcome of a scientific investigation v. outline improvements or extensions to the method that would benefit the scientific investigation.	i. Raw, quantitative data from multiple trials are neatly presented in a table with headings and relevant units. Averages calculated and presented in a table. A correct type of graph is plotted. The graph has correctly labelled axes including units and suitable scales. Qualitative data from the experiment recorded. ii. Trend shown in the graph is correctly outlined. Anomalies/outliers are identified. Trend is explained using correct scientific knowledge relating to the chosen factor and photosynthesis. iii. Correct conclusion is drawn as to how well the data support the hypothesis and answer the research question, with reliability of the data stated. iv. At least two limitations relating to the validity of the investigation are outlined. Consider how the limitations affected the data and how well the research question could be answered. v. At least one improvement relating to each limitation raised is described in detail. Extensions of investigation described.
Level	Level Descriptor	Task Specific Clarification
7-8	The student is able to: i. correctly collect, organize, transform and present data in numerical and/ or visual forms	i. Raw, quantitative data from multiple trials are neatly presented in a table with informative headings and relevant units. Rate of photosynthesis and averages correctly calculated and presented in a table.

	ii. accurately interpret data and describe results using correct scientific reasoning iii. discuss the validity of a hypothesis based on the outcome of a scientific investigation iv. discuss the validity of the method based on the outcome of a scientific investigation v. describe improvements or extensions to the method that would benefit the scientific investigation.	A correct type of graph is accurately plotted. The graph has correctly labelled axes including units and suitable scales. Meaningful qualitative data taken throughout the experiment recorded. ii. Trend shown in the graph relating to the factor investigated and rate of photosynthesis is correctly outlined with reference to specific data points. Anomalies/outliers are correctly identified. Trend is explained using correct scientific knowledge relating to chosen factor and photosynthesis. iii. Correct conclusion is drawn as to how well the data support the hypothesis and answer the research question, with reliability of the data discussed. iv. At least two strengths and limitations relating to the validity of the investigation are discussed in detail. Consider how the limitations affected the data and how well the research question could be answered. v. At least one improvement relating to each limitation raised is described in detail. Relevant extensions of investigation described.
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Student Task Planning Notes:

MYP Practical Report Checklist - Student

Before you submit your practical report to your teacher you must use the checklist below to ensure all parts of your report have been completed.

	Section	What you need to do	Completed Y/N
DESIGN (MYP Criterion B: Inquiring and Designing)	Title	Give a name to the experiment you are conducting	
	Research Question	The research question should be clear , concise , and include the independent variable and the dependent variable.	
	Background Information	You should give the science theory behind your investigation, and it should include scientific terminology. This should be presented as a short scientific paragraph .	
	Hypothesis	What do you predict will happen in your experiment and why ?	
	Variables	Independent Variable - what will you change and how will you change it? Dependent Variable - what will you measure and how will you measure it? Control Variables - what are they? How and why are they controlled? I predict that as light intensity increases the rate of photosynthesis increases (more oxygen bubbles per minute), because light supplies the energy needed for photosynthesis.	
	Apparatus	Have you completed a diagram of assembled equipment that shows how to successfully collect data? Have you used a ruler and pencil to do this? Have you labeled all the chemicals and equipment in your diagram?	
	Method	Have you written your method in the following format: - Stepwise - Logical order of steps in investigation - Used correct names for chemical and equipment - Provided detailed account of the steps in the method - Describe how you will change the independent variable - Describe how control variables are monitored	

Change the light intensity by moving the lamp to set distances from the pondweed (10, 20, 30, 40 cm) and measure each distance with a ruler.

	Safety	Discuss any safety issues with chemicals or equipment being used, and describe what precautions should be taken to ensure you conduct the investigation safely.	
RESULTS and CONCLUSION (MYP Criterion C: Processing and Evaluating)	Results Table	A table of results must include: - Headings - Units - Repeats - Averages	
	Observations	Record detailed observations: - At start of experiment - During experiment - At end of experiment Observations can include colour change, bubbles, heat etc.	
	Calculations	Calculate averages of repeats. Show formula for calculation Show working out for calculations Give final answer to calculation with correct units .	
	Graph	Draw a graph to include the following: - Heading, labels with units on each axis - Suitable scales that occupy minimum $\frac{3}{4}$ of each axis - Plot all points accurately - Draw a curve or straight line of best fit that ignores points that do not fit the pattern. - Discuss trends or patterns observed in graph.	
	Conclusion	Summarize what your results show. Does this support your prediction/hypothesis ? Explain your results using scientific knowledge ?	
	Evaluation	Discuss the reliability of your results . Discuss the validity of the method you used.	
	Improvements	Suggest improvements to the scientific method and describe how they can reduce or eliminate errors	

Yes - the results support my prediction: as light intensity increased, the number of oxygen bubbles per minute increased, because light provides the energy for photosynthesis.

8B 9a Respiration and Photosynthesis

1 Copy and complete these word equations.

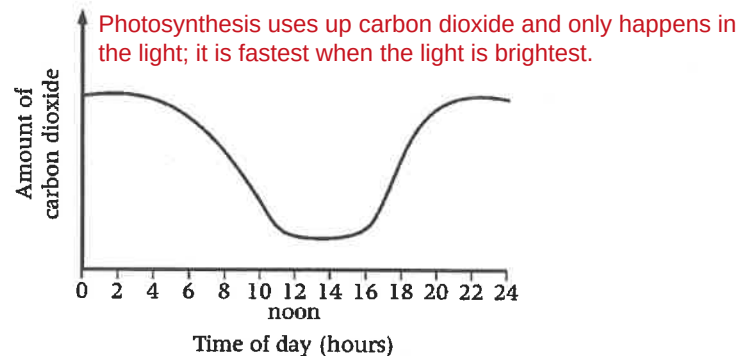
a Photosynthesis:

carbon dioxide + water (+ light energy) → glucose + oxygen

b Aerobic respiration:

oxygen + glucose → carbon dioxide + water (+ energy)

2 Scientists can find out about photosynthesis by measuring the concentration of different gases near plants. This graph shows the concentration of carbon dioxide in the air around an oak tree during a day in May.



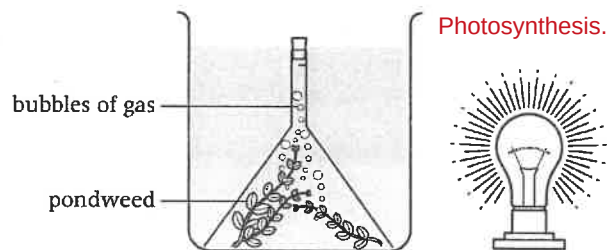
a Which process in a plant uses up carbon dioxide?

b When do you think carbon dioxide was being used up by the tree?

c At which times of the day do you think that the light was brightest? Explain your answer.

d What does this tell you about the process that uses up carbon dioxide?

3 It is known that photosynthesis gets faster if there is more light. It is also known that photosynthesis gets faster if there is more carbon dioxide. This experiment was set up to see if this was also true for plants that grow under water.



a What gas did the bubbles contain? Oxygen (given off during photosynthesis).

Write down whether each of the following changes would make the number of bubbles go up or down. Explain your reasoning.

b The light bulb was moved further away.

c Why do you think this?

d Extra carbon dioxide was dissolved in the water.

e Why do you think this?

The number of bubbles would go down: moving the lamp further away reduces the light intensity, so photosynthesis slows down and less oxygen is produced.

Because light intensity decreases as the distance from the lamp increases, and photosynthesis needs light; less light means a slower rate.

The number of bubbles would go up: extra carbon dioxide makes photosynthesis faster, so more oxygen is produced.

Because carbon dioxide is a raw material of photosynthesis; the more CO₂ available, the faster the reaction can go (up to a limit).

What do you think today's lesson is going to be about?

What could today's learning objectives be? (use command words such as 'describe', 'explain' etc)

About photosynthesis and respiration - how plants make food using light and how living things release energy from food.



Describe the word equations for photosynthesis and aerobic respiration; explain the factors that affect the rate of photosynthesis.

8B 10a Biofuels Information Sheet



Biofuels have been around as long as cars have. At the start of the 20th century, Henry Ford planned to fuel his Model Ts with ethanol, and early diesel engines were shown to run on peanut oil.

But discoveries of huge petroleum deposits kept gasoline and diesel cheap for decades, and biofuels were largely forgotten. However, with the recent rise in oil prices, along with growing concern about global warming caused by carbon dioxide emissions, biofuels have been regaining popularity.

Gasoline and diesel are actually ancient biofuels. But they are known as fossil fuels because they are made from decomposed plants and animals that have been buried in the ground for millions of years. Biofuels are similar, except that they're made from plants grown today.

Much of the gasoline in the United States is blended with a biofuel—ethanol. This is the same stuff as in alcoholic drinks, except that it's made from corn that has been heavily processed. There are various ways of making biofuels, but they generally use chemical reactions, fermentation, and heat to break down the starches, sugars, and other molecules in plants. The leftover products are then refined to produce a fuel that cars can use.

Countries around the world are using various kinds of biofuels. For decades, Brazil has turned sugarcane into ethanol, and some cars there can run on pure ethanol rather than as additive to fossil fuels. And biodiesel—a diesel-like fuel commonly made from palm oil—is generally available in Europe.

On the face of it, biofuels look like a great solution. Cars are a major source of atmospheric carbon dioxide, the main greenhouse gas that causes global warming. But since plants absorb carbon dioxide as they grow, crops grown for biofuels should suck up about as much carbon dioxide as comes out of the tailpipes of cars that burn these fuels. And unlike underground oil reserves, biofuels are a renewable resource since we can always grow more crops to turn into fuel.

Unfortunately, it's not so simple. The process of growing the crops, making fertilizers and pesticides, and processing the plants into fuel consumes a lot of energy. It's so much energy that there is debate about whether ethanol from corn actually provides more energy than is required to grow and process it. Also, because much of the energy used in production comes from coal and natural gas, biofuels don't replace as much oil as they use.

For the future, many think a better way of making biofuels will be from grasses and saplings, which contain more cellulose. Cellulose is the tough material that makes up plants' cell walls, and most of the weight of a plant is cellulose. If cellulose can be turned into biofuel, it could be more efficient than current biofuels, and emit less carbon dioxide.

8B 10b Plus – Minus - Interesting

Use the biofuel information sheet to complete the following table:

PLUS MINUS INTERESTING (PMI)	
From 2025 the only source of fuel will be biodiesel made from vegetable oil	
Benefits (plus) PLUS: Renewable fuel - we can grow more crops, and plants absorb carbon dioxide as they grow, so burning it adds less CO₂ to the atmosphere. MINUS: Takes land away from growing food, may reduce biodiversity, and lots of energy (fertilisers, processing) is needed to make it.	Harms (minus)
Interesting thoughts or ideas	

INTERESTING: Henry Ford ran Model T cars on ethanol and early diesel engines ran on peanut oil - biofuels are not a new idea.

10c Renewable Energy Resources – Is biodiesel the Answer?

Virtually everyone thinks that renewable sources of energy are better than continuing to burn fossil fuels. However, people can't agree on which renewable energy source would be the best choice.

Biodiesel seems to be a good choice. It is made from plants which use up carbon dioxide during **photosynthesis**. So, when it is burned, it will not add to the 'greenhouse effect'.

However, many people are worried about losing land needed for growing food and how this affects **biodiversity**.

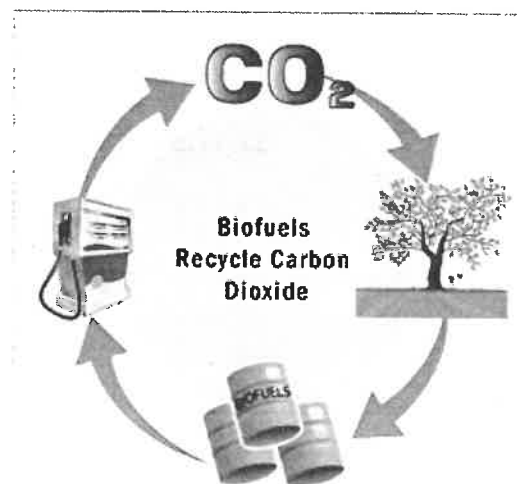
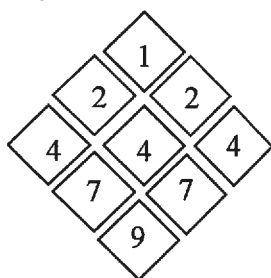


Image with permission from
http://community.middlebury.edu/~cri/biodiesel_cycle.png

Instructions:

Collect a set of 12 biodiesel cards for your discussion group.

You must arrange them into a 'Diamond 9'. You must decide which nine contains the most important information so these can be arranged as follows:



Most convincing
reason for using



Least convincing
reason for using



This will mean discarding three of the cards. If you wish, you may discard an extra **one or two cards** and write your own reasons for using these instead on the blank cards provided.

Now compare your diamond 9 with those prepared by other groups.

Are they all the same? Can you persuade another group to change the position of any of their cards? Are you willing to change the position of any of your cards?

8B 10d Biofuels Q and A

Once heralded as the alternative fuel source of the future, biofuels have come under scrutiny recently with a number of reports suggesting they cause more harm to the environment than originally thought.



Q1) What are biofuels? Biofuels are fuels made from plants grown today, for example ethanol from corn or sugarcane and biodiesel from vegetable oil.

Q2) Where do we get them from?

From crops grown today such as corn, sugarcane and oil palm; the plant material is broken down using fermentation, chemical reactions and heat.

Q3) Are they a new invention? Explain your answer.

Q4) Why did we stop using biofuels?

No - Henry Ford planned to run his Model T cars on ethanol in the early 20th century, and early diesel engines ran on peanut oil.

Huge deposits of petroleum were discovered, which kept gasoline and diesel cheap, so biofuels were largely forgotten.

Q5) So why is there renewed interest?

Rising oil prices and growing concern about global warming caused by carbon dioxide emissions have made biofuels popular again.

Q6) Are biofuels expensive to buy in the UK? Explain your answer.

Yes - biofuels generally cost more than petrol/diesel because growing and processing the crops uses a lot of energy, and they are made in smaller quantities.

Q7) What are the benefits?

Renewable (we can grow more crops), reduce carbon dioxide emissions because plants absorb CO₂ when growing, and reduce dependence on fossil fuels.

Q8) Are they good for the environment? Explain your answer.

Partly - they release less CO₂ than fossil fuels, but making fertilisers and pesticides and processing the plants uses energy, often from fossil fuels.

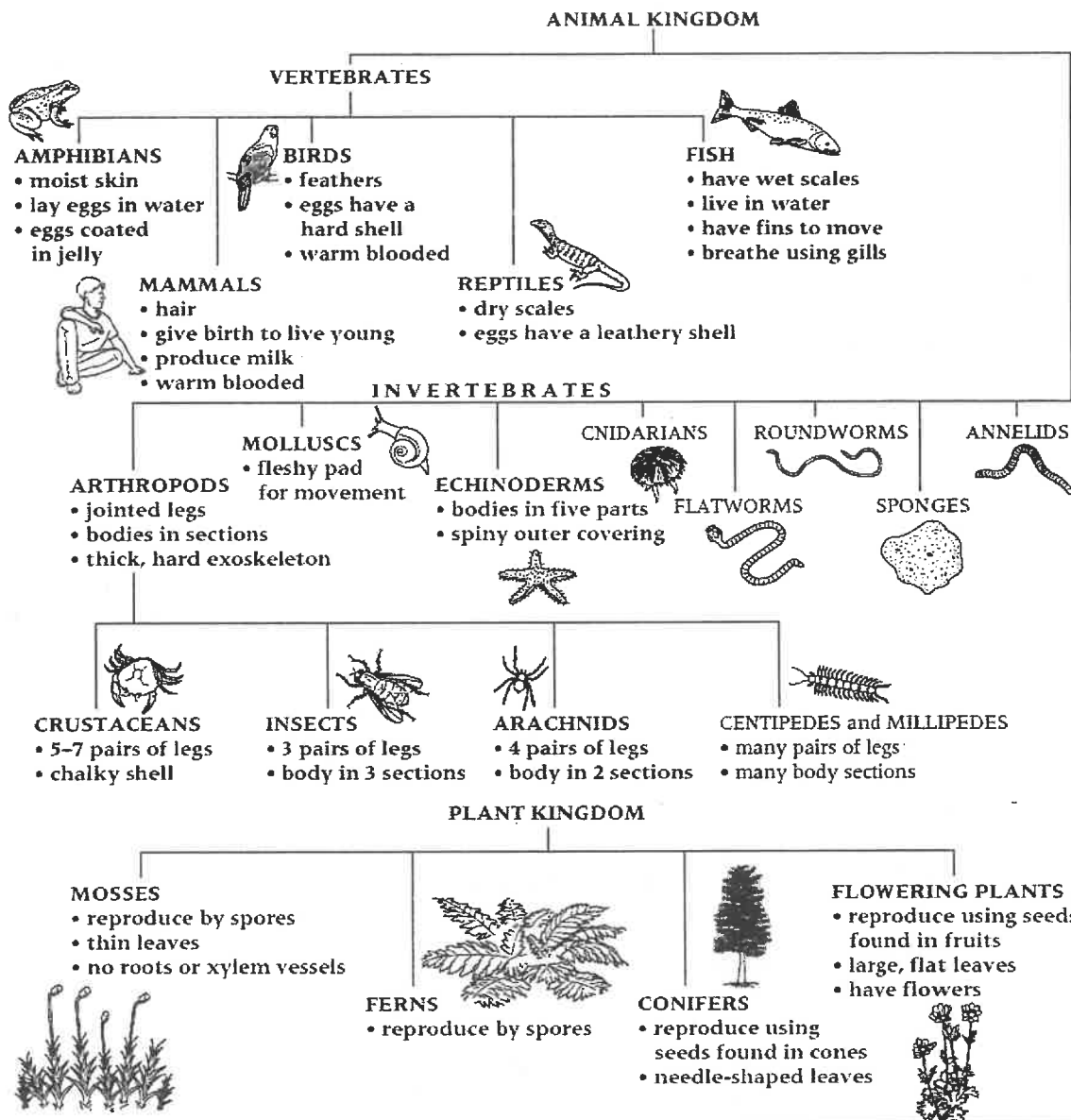
Q9) Are there any negative impacts? Explain your answer.

Yes - land used for fuel crops is lost for growing food, habitats and biodiversity can be harmed, and producing biofuels uses large amounts of energy.

Q10) Are some biofuels better than others? Explain your answer.

Yes - biofuels made from grasses and saplings contain more cellulose, so they could be more efficient and emit less carbon dioxide than corn ethanol.

8B 12a Sorting Kingdoms





MOSSES

- reproduce by spores
- thin leaves
- no roots or xylem vessels



FERNS

- reproduce by spores



CONIFERS

- reproduce using seeds found in cones
- needle-shaped leaves



FLOWERING PLANTS

- reproduce using seeds found in fruits
- large, flat leaves
- have flowers



- 1 Draw a table like the one below.
- 2 Look at the exhibition of organisms.
- 3 Write the name of each animal or plant in your table and decide which groups it is in. Use the diagram above to help you. For example, an ant is an animal, an arthropod and an insect.
- 4 Write down the reason why you have put it in the smallest of these groups. For example, an ant is an insect because it has six jointed legs.

Name	Kingdom	Group(s)	Reason
ant	animal	arthropod; insect	has six jointed legs

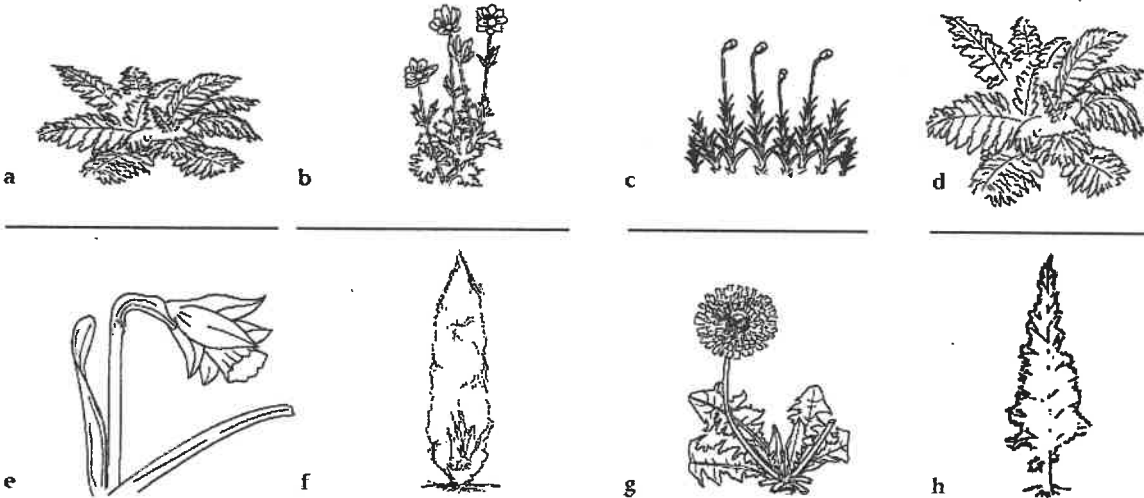
• Draw a table with columns: Name | Kingdom | Group(s) | Reason, and complete it for each organism from the exhibition.

Example row: Ant | Animal kingdom | Arthropod, insect | It has six jointed legs and a body in three sections.

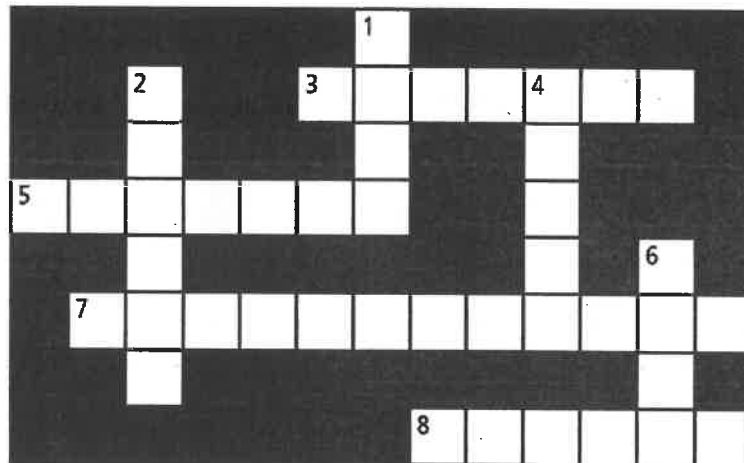
8B 12b Identifying Plants



1 Look carefully at these pictures. Write the words conifer, fern, flowering plant or moss underneath the correct picture.



2 Using the clues below, fill in the answers on this crossword.



Across

3 A fir tree is in this plant group.

5 Things made by some plants which are often brightly coloured.

7 Seeds and spores are used for this.

8 Conifers have needle-shaped ones.

Down

1 Group of small plants that make spores.

2 Ferns reproduce using these.

4 Contains seeds from a flowering plant.

6 Contains seeds from a conifer.



literacy, knowledge

Label the pictures: flowering plant (flowers, seeds inside fruits), conifer (needle-shaped leaves, seeds in cones), fern (large leaves, reproduces by spores), moss (thin leaves, no roots, reproduces by spores).

Across 3: conifers - a fir tree is a conifer.

Down 2: spores - ferns reproduce using these.

Down 4: fruit - contains seeds from a flowering plant.

Down 6: cone - contains seeds from a conifer.

Across 8: leaves - conifers have needle-shaped ones.

8B 12c Grouping Plants



1 Using the words in the box, fill in the correct word in each of the blank spaces.

a _____ and _____

both use seeds for reproduction.

Conifers

_____ and _____

both use _____ for reproduction.

b _____ contain seeds in flowering plants.

c _____ contain seeds in conifers. Flowering plants

d Mosses have no xylem vessels.

e Xylem vessels carry _____

around the plant. xylem vessels

f Plants make their own food using a process called photosynthesis.

g The life process that plants use to release energy

from food is called respiration.

flowering plants

2 Find out the name of one plant in each group. Write the name of the plant and the group that it is in.

Name of plant	Plant group that it belongs to
Conifers	
cones	
	flowers

cones

conifers

ferns

flowering plants

fruits

mosses

photosynthesis

respiration

spores

water



literacy, knowledge, research

spores

water

• Question 2 - research task: name one plant from each group and say which group it belongs to. Example answers given in the table below.

Conifer - Scots pine (*Pinus sylvestris*)

Fern - male fern (*Dryopteris filix-mas*)

8B 13a Using Identification Keys

1. The diagram shows an animal.



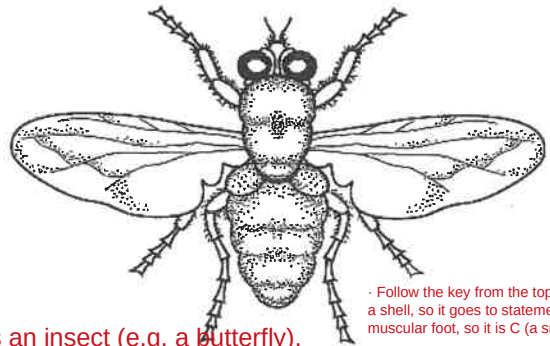
Use the key to identify the animal.

- 1) has legs go to 2
has no legs go to 3
 - 2) has a shell *animal A*
has no shell *animal B*
 - 3) has one muscular foot *animal C*
has more than one foot *animal D*
- it has a shell and one muscular foot (a snail).*

2. The diagram shows an animal.

Use the key to identify the animal.

1. wings present go to 2
wings absent go to 3
2. one pair of wings visible A
two pairs of wings visible B
3. three pairs of legs C
four pairs of legs D



Follow the key from the top: the animal shown has a shell, so it goes to statement 2, and it has one muscular foot, so it is C (a snail).

3. The table shows some features of animals.

Which animal could be a bird?

The bird is the animal with wings present and legs present; only one row (B) has x in both columns.

	feature			
	feathers	gills	hair	wings
A	x	✓	✓	x
B	✓	✓	x	x
C	✓	x	x	✓
D	x	x	✓	✓

Follow the key: wings present, then two pairs of wings visible so the animal is B (an insect such as a butterfly).

key

✓ = present

x = not present

B - a bird has both legs and wings present, so it must be the animal with x in both the legs and the wings column.

This page is a picture/diagram plate (key diagrams and photographs); it carries no written questions - no written answers are needed, the drawings are used with the identification keys.

4. The photos shows six members of the cat family, Felidae

Use the key to identify the cats, A to F.

Tick () the boxes in Table 1.1 to show how you identify each cat.
Write the name of each cat in the correct box in Table 1.1.

Cat A has been completed for you as an example.

Question	Name of cat
1 a Ears pointed b ears rounded	Go to 5 Go to 2
2 a Fur with no stripes b Fur with stripes or spots	P. leo Go to 3
3 a Fur with stripes, but no spots b Fur with spots	P. tigris Go to 4
4 a Fur with spots, but no stripes b Fur with spots and stripes	A. jubatus N. nebulosi
5 a Fur with spots b fur with no spots	L. rufus L. caracal



A



B



C



D



E



F

Table 1.1

cat	1 (a)	1 (b)	2 (a)	2 (b)	3 (a)	3 (b)	4 (a)	4 (b)	5 (a)	5 (b)	name of cat
A											L.caracal
B											A.Jubatus
C											P. leo
D											N.nebulosi
E											L. rufus
F											P. tigris

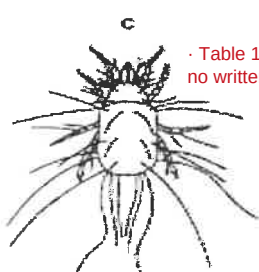
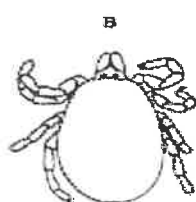
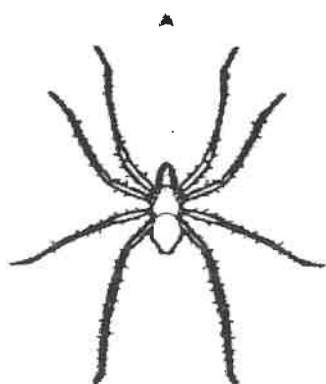
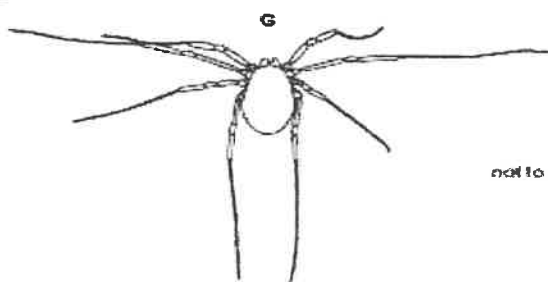
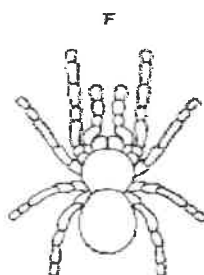
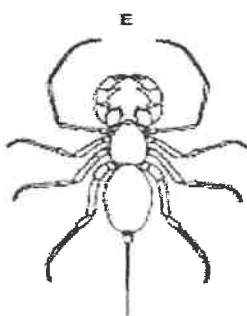
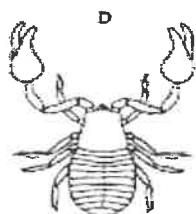


Table 1.1 and the drawings (A, B, C) are picture plates for the classification-key work. no written answers needed - they are used with the arachnid key on page 65.



not to scale

Extension

5. The diagrams show seven species of arachnid.

Use the key to identify each species.

Write the letter of each species (A to G) in the correct box beside the key.

One has been done for you.

Key

1 (a)	Tail present	<i>Abaliella dicranotarsalis</i>	
(b)	Tail not present	go to 2	
2 (a)	Legs much longer than body	go to 3	
(b)	Legs not much longer than body	go to 4	
3 (a)	Hairs on the legs	<i>Tegenaria domestica</i>	
(b)	No hairs on the legs	<i>Odielus spinosus</i>	
4 (a)	Body segmented	<i>Chelifer tuberculatus</i>	
(b)	Body not segmented	go to 5	
5 (a)	2 body parts about the same size	<i>Poecilotheria regalis</i>	
(b)	One part of the body larger than the other	go to 6	
6 (a)	Body covered in long hairs	<i>Tyroglyphus longior</i>	
(b)	Body not covered in hairs	<i>Ixodes hexagonus</i>	

· Write the letter of each species in the box beside the key statement. Letters A-G below are given in key order, with the feature that identifies each species.

A - *Abaliella dicranotarsalis* (tail present) - already done for you

Poecilotheria regalis (two body parts about the same size)

Tyroglyphus longior (body covered in long hairs)

Ixodes hexagonus (body not covered in hairs)

B - *Tegenaria domestica* (hairs on the legs)

C - *Odielus spinosus* (no hairs on the legs)

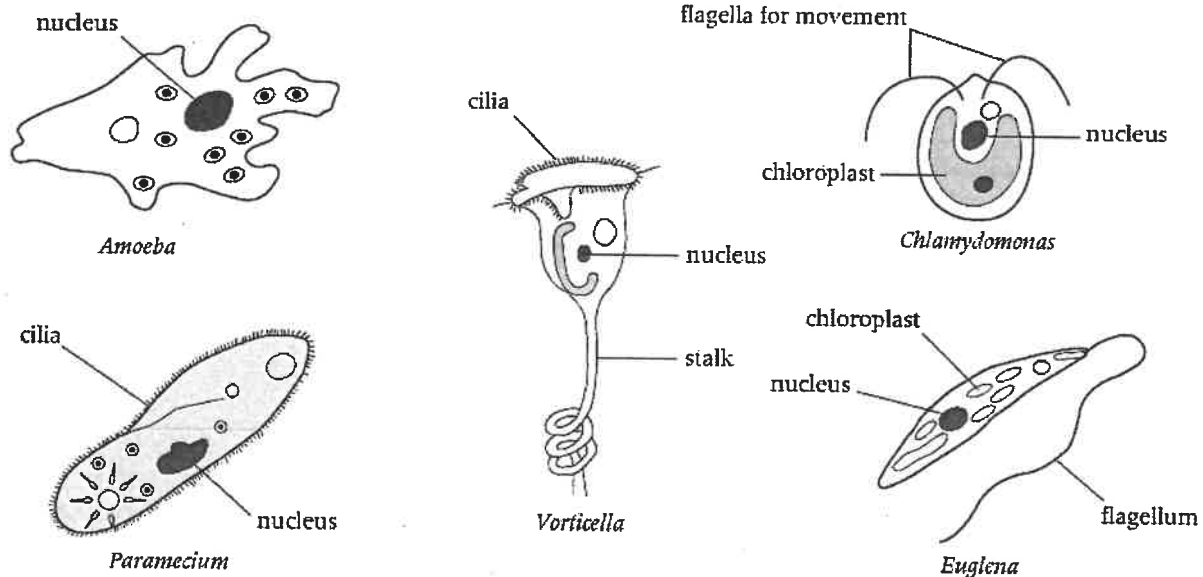
D - *Chelifer tuberculatus* (body segmented)

8B 13b Constructing Keys

Here are some drawings of simple organisms that are found living in water.

Each one is a **unicellular** organism. This means that it is made up of just one cell.

In order to see them you would have to use a microscope.



Look carefully at the labelled features of these organisms. Using these features design a statement key to classify them.

Your key should allow someone else to find the names of the organisms if they were just given the labelled drawings.

1 Has chloroplasts (green) -> Chlamydomonas, Euglena.
 2 Has cilia -> Paramecium; has a stalk -> Vorticella; no fixed shape and moves with pseudopodia -> Amoeba.



8B 13c Make your own Classification Key

1. Choose 4 different animals online.

Choose animals that are similar. For example:

- 4 different dogs,
- 4 different cetaceans (whales or dolphins)
- 4 different sharks,
- 4 different rodents
- 4 different insects, etc.....

2. Paste a picture of each of these animals in this table.

Include their common name and their scientific (bi-nomial) name.

Model answers use four similar animals (big cats). Any four closely related animals with correct common and scientific (binomial) names are acceptable.

Animal 1: Common name: Animal 2 - Common name: tiger; Scientific name: Panthera tigris Scientific name:	Animal 2: Common name: Scientific name:
Animal 3: Animal 3 - Common name: jaguard; Scientific name: Panthera pardus Common name: Scientific name:	Animal 4: Common name: Scientific name:

3. design a statement key to classify them. Again, your key should allow someone else to find the names of the organisms if they were only given your pictures to look at.

Submit on the GC when you are finished

Write your statement key here:

My Key:

1.

3 Has large rosette spots -> leopard; has smaller rosettes with a spot in the middle -> jaguar.

8B 14a Constructing a Savannah Food Web

Ecologists have been studying the feeding relationships of animals in a savanna grassland. They want to present these relationships in the form of a **food web**.

Pictures of these animals and plants, together with their common names and scientific names, are found on the next page.

1. Cut out pictures of the animals together with their names.
2. Read the sentences below.
3. Complete a **food web** by joining the connections with the pictures. Remember the arrows should show one organism being "eaten by" the other.

• Arrows must point in the direction of energy flow - from the organism that is eaten to the organism that eats it ('eaten by').

Feeding relationships:

- Rhinoceros *feed on* grass
- Rhinoceros and giraffes *eat* the leaves of **Acacia trees**
- Grasshoppers and mice *compete for* grass
- snakes eat mice
- Ocelots *feed on* snakes
- Vultures *feed on* dead mongooses, **leopards** and rhinoceros
- Monkeys *feed on* grasshopper and Snakes
- Ocelots *feed on* mice
- Impala *feed on* **small shrubs** and grass
- Mongooses *feed on* drongo birds
- Monkeys and drongo birds *compete for* grasshoppers
- Lions *feed on* rhinoceros and giraffes
- Leopards *feed on* impalas and monkeys

• Completed food web: grass -> grasshopper -> monkey -> leopard;
 grass -> mouse -> snake -> ocelot;
 grass -> rhinoceros -> lion; Acacia leaves -> giraffe/rhinoceros -> lion;
 grasshoppers -> drongo birds -> mongooses; Acacia leaves/shrubs/grass -> impala -> leopard;
 dead leopards, mongooses and rhinoceros -> vultures.



Snake

Python sebae



Mongoose

Herpestes vitticollis



Monkey

Chlorocebus pygerythrus



Leopard

Panthera pardus



Lion

Panthera leo



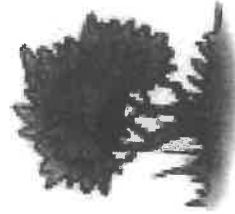
Grasshopper

Hieroglyphus daganensis



Mouse

Rhabdomys dilectus



Small Shrubs

Salvadora persica

· Picture page (savannah animals and plants) to be cut out for the food-web activity on page 68; no written answers.



Giraffa giraffa



Vulture

Gyps africanus



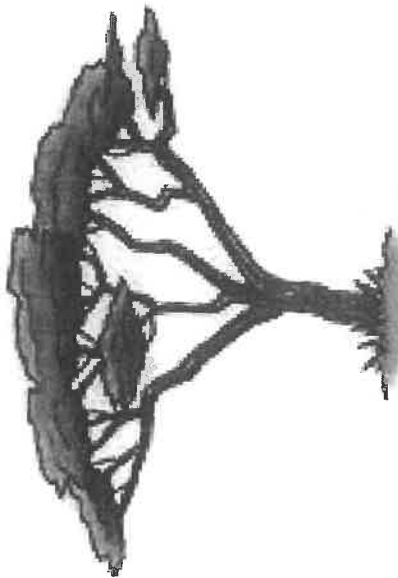
Grass

Themeda triandra



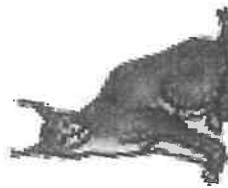
Dicrurus macrocercus

· Picture/diagram page (savannah organisms); no written questions - no written answers needed.



Acacia tree

Acacia caffra



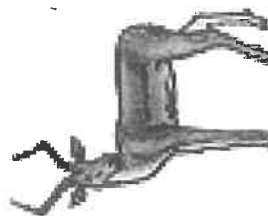
Ocelot

Leopardus pardalis



Rhinoceros

Ceratotherium simum



Impala

Aepyceros melampus

8B 14b Chains, Webs and Pyramids

Here are three food chains:

- A grass → rabbit → fox
 B wheat → field mouse → fox
 C wheat → field mouse → grass snake



1 Sketch a pyramid of numbers for food chain A.

Pyramid of numbers for chain A (grass → rabbit → fox): very wide base of many grass plants, a narrower middle layer of rabbits, and a tiny top of foxes.

2 Use all three food chains to draw a food web.

Food web joins all three chains: grass and wheat are the producers; rabbits and mice are primary consumers; foxes and grass snakes are secondary consumers.

3 Name one producer in the food web. _____

4 Name a primary consumer. _____

5 Name a secondary consumer. ~~fox (or grass snake)~~ _____

6 a If the field mice died, what would happen to the number of grass snakes?
 grass (or the wheat field) _____

b Why would this happen? ~~grass snakes feed on mice, so with fewer mice there is less food; fewer grass snakes can survive and breed.~~
 rabbit (or mouse) _____

c If the field mice died, what might happen to the number of rabbits?
 The number of grass snakes would decrease, because their food supply has disappeared. _____

d Why might this happen?
 The number of rabbits might increase, because there is less competition for the plant food. _____



knowledge

With fewer mice eating the wheat and grass, more plant food is left for the rabbits, so more rabbits survive and reproduce.

8B 14c Pyramids of Numbers



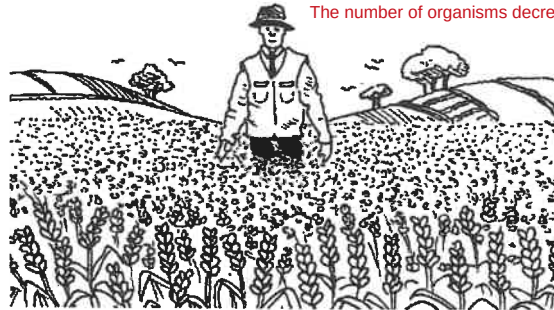
1 Sketch a pyramid of numbers for each of the following food chains:

- a grass → rabbit → fox
- b oak tree → caterpillar → robin → owl
- c lettuce → slug → thrush → cat → flea



Farmer Giles.

· Chain a: grass → rabbit → fox.
Pyramid of numbers: many grass plants at the base, fewer rabbits, fewest foxes at the top.



Farmer Green.

The number of organisms decreases as you go up the pyramid.

Farmer Giles reared chickens, which he then sold for humans to eat. He worked out that to feed 1 human he would have to produce 40 chickens. To feed the 40 chickens he had to grow 10 000 corn plants.

Farmer Green just grew corn. He sold his corn to be fed to humans. He worked out that he needed to grow 1000 corn plants to feed 1 human.

· Chain c: lettuce → slug → thrush. Pyramid of numbers: many lettuce plants at the base, fewer slugs, fewest thrushes at the top.



2 a Draw the food chain for Farmer Giles' farm. corn → chicken → human

b Draw a pyramid of numbers for this food chain.

3 a Draw the food chain for Farmer Green's farm.

b Draw a pyramid of numbers for this food chain.

4 a What happens to the number of organisms as you go up each pyramid?

b What happens to the size of the organisms as you go up each pyramid?

5 a How much more land than Farmer Green would Farmer Giles need to feed 1 human? Ten times more (10000 / 1000 = 10), so Farmer Green needs ten times as many corn plants.

b Why is it more efficient to produce food for humans from plants than from animals? Energy is lost as heat at every stage of a food chain, so feeding plants straight to people wastes far less energy and needs less land.



knowledge

· Chain b: oak tree → caterpillar → robin → owl. Pyramid of numbers: one oak tree at the base with many caterpillars above it, so the pyramid is inverted (narrow base, wide second layer).



The organisms get larger in size as you go up the pyramid.

· Extra chain: cat → flea. Pyramid of numbers: one cat at the base with many fleas above it (an inverted pyramid).

· Pyramid of numbers: 10000 corn plants at the base, then 40 chickens, then 1 human at the top.

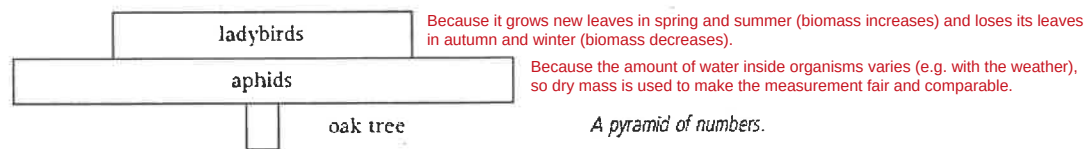
corn → human

· Pyramid of numbers: 1000 corn plants at the base and 1 human at the top.

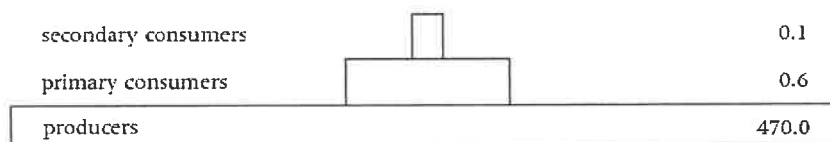
8B 14d Pyramids of Biomass

Each level on a **pyramid of numbers** is called a **trophic level**. Animals that are on the same level in two different pyramids of numbers are said to be at the same trophic level. Producers, for example, are always at Trophic level 1.

Pyramids of numbers do not take into account the *size* of the organisms. A better way of showing things is to use a **pyramid of biomass**. This shows the actual *mass* of material at each trophic level. A sample is taken and the masses of the different organisms are measured. This often involves drying the organisms in an oven to get rid of all the water. How wet or dry the weather is can affect how much water is found inside an organism. This would affect the **biomass** and might make the measurement inaccurate. So, the biomass is a measure of the *dry mass* of material at each trophic level.



- 1 Explain why the pyramid of numbers above is not shaped like a pyramid.
- 2 What number trophic level are the ladybirds at? *trophic level 3 - the ladybirds are secondary consumers.*
- 3 Draw a food chain for this pyramid of numbers. *oak tree -> aphid (greenfly) -> ladybird*
- 4 What do you think the word 'biomass' means? *the mass of living material (the dry mass of the organisms) at each trophic level.*
- 5 Sketch a pyramid of biomass from the pyramid of numbers above. You do not need to add any figures, just draw the shape.
- 6 To work out a pyramid of biomass, the dry masses of the organisms are used. Why?
- 7 The biomass of an oak tree changes during the course of a year. Why is this?
- 8 Only about 10% of the chemical energy in an organism's food is used for growth. What happens to the rest? *the rest is lost mostly as heat from respiration, and some in waste (faeces and urine) and in movement.*
- 9 The biomass at each trophic level gets smaller as you go up the pyramid. Why do you think this is? (*Hint: Think about 'energy loss'.*)
- 10 A difficult question! The diagram below shows a pyramid of biomass for a field. The units are g/m^2 . Describe what this pyramid tells you in as much detail as you can.



Because the producer is a single large oak tree but it supports many thousands of insects, so the base is narrow and the pyramid is inverted. *not to scale*



knowledge, literacy

Pyramid of biomass: a broad base (large mass of oak tree) tapering upwards, because the mass of living material gets smaller at each higher trophic level.

8B 15a Ecological Terms

Community - all the different organisms living in the place.

- 1 Look at the list of words that are often used when studying habitats. Match each word with its meaning by colouring them in the same colour.

community	A person who studies habitats. Ecologist - a person who studies habitats.
ecologist	The place where an organism lives.
environment	The number of organisms of one species living in a habitat.
habitat	How much water vapour there is in the air.
humidity	All the different organisms living in the same place.
light intensity	A sampling square used to estimate plant populations.
microhabitat	The surroundings of an organism.
population	How bright the light is. light intensity - how bright the light is.
quadrat	Small area in a habitat where organisms live.

- 2 There are 150 lions living on the grasslands in one part of Africa. They feed on large animals such as gazelles, zebras and wildebeest and these feed on the leaves of acacia trees and grasses.

Habitat - the place where an organism lives.



- a What is the population of lions? _____
- b What is the community on the grasslands? the grasslands: lions, gazelles, zebras, wildebeest, acacia trees and grasses. _____
- c What is the habitat of the lions? The grasslands of Africa (the area where the lions live). _____

Extension: List factors that will affect the population of lions.

Population - the number of organisms of one species living in a habitat.

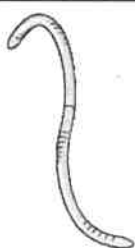
Humidity - how much water vapour there is in the air.

8B 15b Survival Adaptations

The table below gives the names and adaptations of some organisms.

Complete the table, giving a reason for each adaptation. (*Hint:* You may need to find out where some of the organisms live.)

In the last row fill in the information for another organism of your choice.

Organism	Adaptation	Reason
Earthworm 	1 Soft slimy skin 2 Bristles 3 Moves away from light	
Limpet 	1 Sucker at bottom 2 Hard shell	
Water milfoil (a pond weed) 	1 Long floppy stem 2 Leaves only present near surface of water 3 Flowers formed above the surface of the pond	
Rabbit 	1 Large ears 2 Powerful hind legs	



8B 15c A Well Adapted Creature

Here is a list of conditions in an imaginary habitat:

- cold, wet and windy weather
- ground water is frozen
- steep rocky ground
- small animals live in dark caves
- tall trees with smooth bark grow in the mountains. They have berries on their highest branches.



Design an imaginary creature that is adapted to survive in this habitat. Make a large drawing and label all its important features.

You will need to think about:

- what it eats and how it gets water
- how it moves around
- how it is protected from the weather.

Drawing must be labelled: thick fur/blubber, gripping claws, camouflage colouring, sheltered nest in a cave, water source, food source, short ears/limbs.



knowledge, considering

Eats lichen, moss, berries from high branches and small animals in caves; gets water by melting snow/ice or from the food it eats.

Moves using strong stout legs and sharp claws with grip pads to climb steep rocky ground in strong wind; small body size keeps it steady.

Protection: thick oily fur coat and layer of fat to keep warm and keep out wet wind; small ears and short limbs reduce heat loss; white/pale colour for camouflage.

Sustainable	The place in which an organism lives is called a	Habitat	All the plants and animals that live in the same place forms a
Community	The total number of individuals of a certain species in a habitat is called a	Population	Living things and non-living things interact to form an
Ecosystem	The conditions around an organism is called the	Environment	A person who studies the relationship between organisms and their environment is an
Ecologist	Components of an ecosystem that are <u>non-living</u> (e.g. light intensity) are the	Abiotic factors	Components of an ecosystem that are <u>living</u> are the ...
Biotic factors	The features that help an organism to survive in its habitat are	Adaptations	The process carried out by plants to make their own food is
Photosynthesis	What an animal does is known as its	Behaviour	If something is allowed to carry on into the future, it is considered

· Full-page image/figure with no readable answer spaces - no written answers required (continuation of the page 78 creature design).

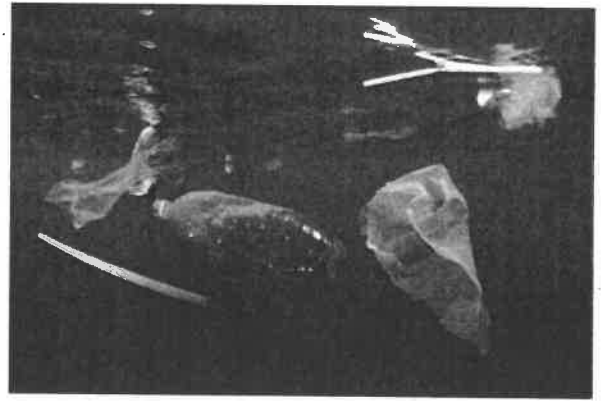
8B 16a A Plastic Ocean

A Plastic Ocean is a documentary made in 2016 to show the devastating impact that the extensive use of plastic has on marine life, and its wider impact on the environment and human populations.

Refuse/reuse plastic bags and bottles, recycle, pick up litter, improve waste collection. This stops plastic reaching the sea, so less is eaten by animals and less enters the food chain.

Website: <https://plasticoceans.org/projects/>

The fish and seafood humans eat contain the built-up microplastics and the toxins attached to them, so humans take in plastic too, which can cause illness and harm health.



Source: <https://gcaptain.com/plastic-waste-in-the-worlds-oceans-could-double-by-2030-iea-warns/>

You will watch a video based on the documentary. Using ideas from the video, your knowledge on food chains and from additional research, answer the following questions.

- 1) What are microplastics and where do they come from?
- 2) How do microplastics enter the food chain?
- 3) What is **bioaccumulation**? You can use the impact of microplastics on the marine food chain to explain this idea.
- 4) What impact does the bioaccumulation of microplastics have on the human population?
- 5) What actions can be taken to reduce this impact? What can you do to help?

Microplastics are pieces of plastic smaller than 5 mm (often too small to see). They come from larger plastic items breaking up in sunlight and waves, and from microbeads and fibres washed out of clothes.

Microplastics are eaten by plankton and small fish, which are then eaten by larger fish and animals, so the plastic passes along the food chain.

Bioaccumulation is the build-up of a substance (microplastics and toxins) in an organism over time, because it is taken in faster than it is removed. It increases up the food chain.

8B 16b A Plastic Ocean Word Search

C	E	Z	B	O	O	P	X	E	D	P	E	P	T	T
G	E	D	C	Z	L	R	S	R	O	W	V	L	E	U
L	Y	E	U	A	V	U	I	L	D	A	I	A	G	R
A	A	R	S	C	E	B	I	S	T	R	T	N	N	T
N	V	T	E	L	A	C	R	A	Q	T	A	K	A	L
D	I	G	G	E	Y	T	S	F	H	S	N	T	H	E
C	U	N	S	U	S	N	I	X	O	T	R	O	C	E
F	I	S	O	L	U	T	I	O	N	S	E	N	V	H
S	N	U	R	D	L	E	B	E	N	X	T	L	N	F
K	N	I	H	T	E	R	L	O	L	M	L	T	I	H
F	P	R	I	V	E	R	S	C	T	A	A	E	H	E
E	S	U	E	R	E	N	M	W	Y	T	H	O	P	A
W	B	L	Y	T	O	K	G	F	N	C	L	W	L	L
N	Z	N	A	V	D	N	X	A	R	N	E	E	O	T
V	W	W	F	I	S	H	J	A	M	W	J	R	D	H

ALTERNATIVE
BOTTLE
CHANGE
CURRENT
DOLPHIN
EDUCATION
FISH

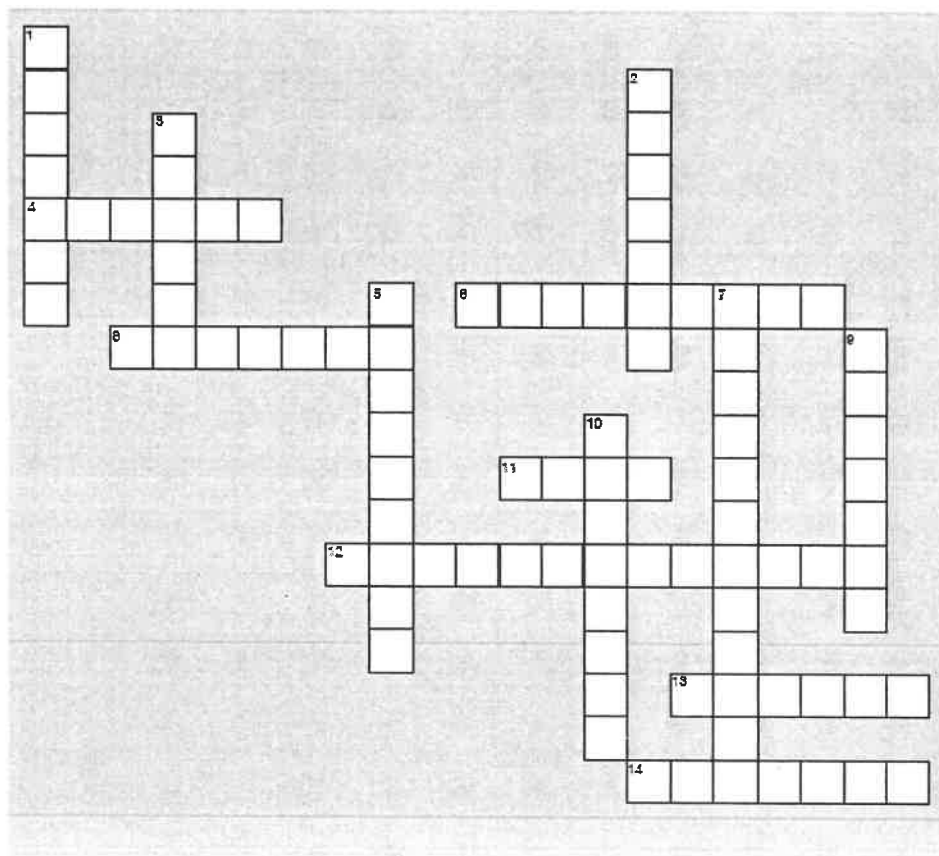
GYRE
HEALTH
LAND
MAGNET
NURDLE
OCEAN
PLANKTON

PLASTIC
POLICY
RECYCLE
RETHINK
REUSE
RIVERS
SEABIRD

SINGLEUSE
STRAW
TOXINS
TURTLE
WATER
WHALE

• Word search: circle all the listed words in the grid (ALTERNATIVE, PLASTIC, GYRE, SINGLE USE, HEALTH, BOTTLE, POLICY, STRAW, LAND, CHANGE, RECYCLE, TOXINS, CURRENT, MAGNET, RETHINK, TURTLE, DOLPHIN, NURDLE, REUSE, WATER, OCEAN, EDUCATION, RIVERS, WHALE, FISH, PLANKTON, SEABIRD).

8B 16c A Plastic Ocean Crossword



Word List

AWARENESS
 BIOACCUMULATE
 CHANGE
 ENTANGLEMENT
 FOOD
 JELLYFISH
 MINUTES
 NURDLES
 PLASTIC
 REFUSE
 RECYCLE
 RETHINK
 SOLUTION
 TUVALU

Across

- 4 The island paradise drowning in plastic
 6 Behavioural change starts with this
 8 The last Resort in the 3 R's **RECYCLE**
 11 1 in 3 sea birds mistake plastic for this
 12 For toxins to travel up the food chain
 13 The first choice to make in the 3 R's
 14 What we want you to do when it comes to your plastic habits

Down

- 1 80% of marine litter is made up of this
 2 Single Use Plastics have an average lifetime of just 12 of these.
 3 We are creating "a wave of"
 5 A turtle's favorite food **JELLYFISH**
 7 A significant source of injury to marine life
 9 Name of the raw pre-production plastic pellets made from oil
 10 We want to be part of this and not the problem

- Crossword grid is only partly legible on the scan.
 Clues read: 4 across island drowning in plastic; 6 across behavioural change starts with AWARENESS; 8 across RECYCLE; 14 across RETHINK; 2 down 12 MINUTES; 3 down CHANGE; 9 down NURDLE.

PLASTIC
 AWARENESS
 MINUTES
 CHANGE
 FOOD
 ENTANGLEMENT
 REFUSE
 NURDLE
 RETHINK
 SOLUTION
 POLLUTION

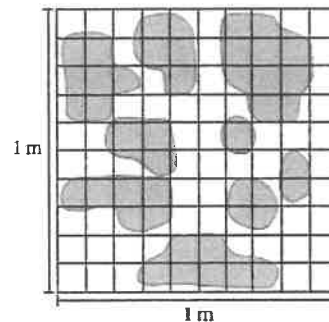
PlasticOceans.org

8B 17a Quadrat Studies

If you wanted to find out how much grass was growing on a field, it would be impossible to count the plants because there would be too many. Instead you can estimate the area of the field covered by the grass.

To do this, you would use a **grid**. This is a quadrat that is divided into 100 small squares. You count the number of small squares that the grass fills. For partly filled squares you ignore those that are less than half filled, but count those that are more than half filled.

If 20 squares out of 100 are filled the **percentage cover** is 20%.



b Plant A is most affected - it has far fewer plants in the frequently walked-on area than in the rarely walked-on area.

c The least affected is the plant whose numbers are almost the same in both areas (e.g. plant E, which is not changed by trampling).



- 1 Look at the grid above. The shaded areas represent patches of grass. Work out the percentage cover of grass in this grid.
- 2 If you were using this technique to estimate the grass cover in a large field, what would you have to do to improve the accuracy of your estimate?

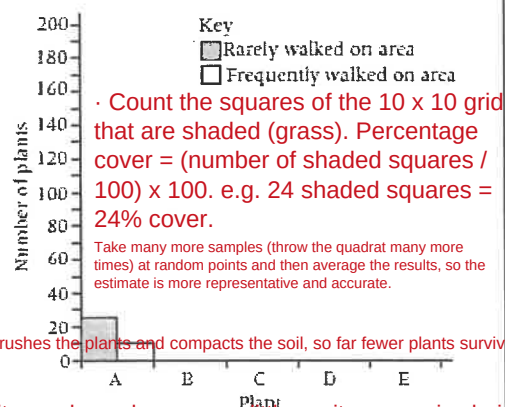
Some students studied two grassy areas. One area was frequently walked over and used as a playing field. The other was rarely walked over.

The students estimated the number of five different plants growing in the two areas using a quadrat to see if they were affected by being walked on. The results are shown in the table below.

Plant	Number in rarely walked on area	Number in frequently walked on area
A	25	10
B	195	160
C	100	5
D	20	75
E	40	40



- 3 a Draw a bar chart to show these results. A bar chart has been started here to show you how to do this. You will need to use a key to show the differences between the bars.
- b Which plant is most affected by being walked over?
- c Which plant is least affected by being walked over?
- d How is plant A affected by being walked over?
- e How is plant D affected by being walked over?



knowledge, presenting

Bar chart: plant A-E along the x-axis, number of plants on the y-axis, two bars per plant, with a key for 'rarely walked on' and 'frequently walked on'.



8B 17b Quadrat Sampling Practical

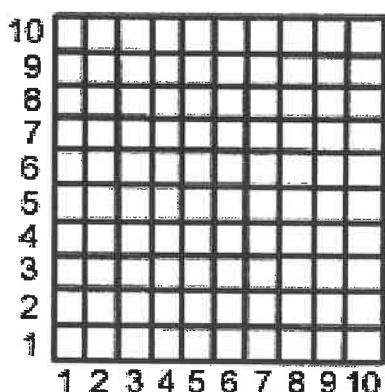
Some mysterious plants have landed on the Island School playground. It has been decided that we should only investigate further if these plants cover more than 20% of the surface of the playground.

Your task is to apply random sampling to estimate the % coverage of these plants on our playgrounds.

Method

1. A 10m x 10m area will be marked out in the playground. The area will be divided into 100 squares shown as the diagram below.
2. You will need to sample 10 squares, determined using a random method.
3. Use this random number generator website to generate 10 pairs of numbers between 1 and 10 <http://www.random.org/integers/>. Do this twice so you have a set of X and Y coordinates.
4. Using the grid below, shade in the 10 squares that you will be sampling in (first number on x axis and second number on y). Record the pairs of numbers.

10 pairs of numbers



X	Y

a $0.5 \text{ m} \times 0.5 \text{ m} = 0.25 \text{ m}^2$

b $10 \times 0.25 \text{ m}^2 = 2.5 \text{ m}^2$ sampled in total.

c Yes - the average % cover per quadrat is an estimate of the % cover of the whole playground, because the squares were chosen at random.

5. At the playground, in each of the squares that you have randomly chosen to sample, place the quadrat at the centre.
6. Count the number of small squares in the quadrat with the plants present (if 50% or more of the square is covered then count it). Since there are 100 squares in your quadrat, this should give you a number out of a hundred and therefore the percentage.
7. Add the 10 numbers together and divide by 10 to calculate the **AVERAGE % COVER per quadrat**.
8. To estimate the % coverage on the playground, consider the following:
 - a. What is the area of one quadrat? (Hint: Each side is 0.5m)
 - b. What is the total area you have sampled using the quadrats?
 - c. Can you calculate the total % coverage?

Answer the following questions

1. In your opinion, how reliable are your results?

2. How can you increase the reliability of your data?

1 Fairly reliable, because 10 squares were chosen at random, but only a small part of the playground was sampled, so the estimate is not exact.

3. List the things that you could do to make the results more accurate.

2 Sample more squares (bigger sample) and repeat the sampling several times, then average the results; use random numbers so the sample is unbiased.

Notes:

3 Use the same sized quadrat each time, place it accurately in the centre of each chosen square, count a square only when 50% or more is covered, and have another person check the counts.

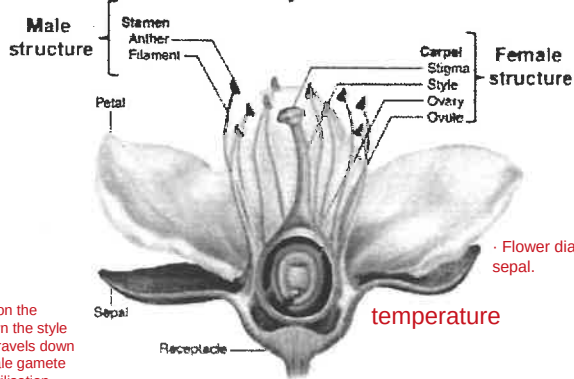
8B 18a Revision of Key Concepts

Read through and complete:

1. Plant organs and reproduction

- The flower contains sex organs **male** and **female**

Parts of an insect pollinated flower



Pollination
anther
stigma
ovary
male gamete
Fertilisation
ovary

Q: Pollen from the anther lands on the stigma. A pollen tube grows down the style to the ovary. The male gamete travels down the tube and fuses with the female gamete (egg cell), forming a zygote - fertilisation.

temperature

- P..... is the transfer of **pollen** from the a..... to the S.....
- A **pollen grain** develops a **pollen tube** to reach the **ovary**
- The n..... from the pollen cell travels down the **pollen tube**.
- F..... takes place in the **ovary** where the n..... from the **pollen cell** fuses with the n..... **gamete** from the **egg cell** **egg cell**
- This develops into an **embryo** which becomes a **seed**
- The **ovary** becomes the **fruit** surrounding the seed.

Q: Cover up the above information and describe how a flower is fertilized using the words **anther, stigma, ovary, pollen, pollen tube, gametes, zygote**

2. Seed germination

- S..... contain **embryonic plants**
- Seeds need the correct t....., oxygen and w..... to **germinate**
- Germination can be investigated by varying these factors
- When a seed germinates, the seedling emerges from the seed coat.

water
oxygen

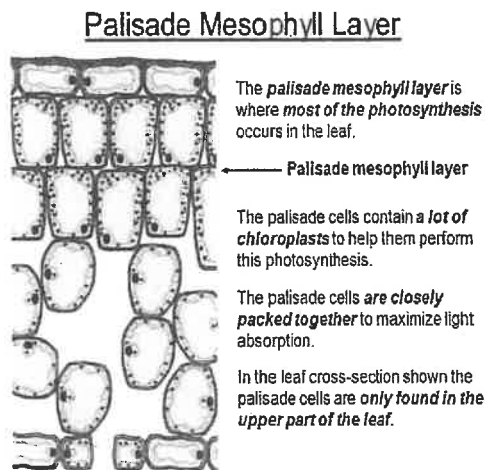
3. Plants make their own food during p.....

- Word Equation

c..... d..... + Water → oxygen + g.....

- L..... is needed for this process. Green plant cells contain a lot of **chloroplasts** which contain a chemical called c..... **chlorophyll**..... to absorb the l.....

Palisade cells in the leaf are adapted for efficient photosynthesis:

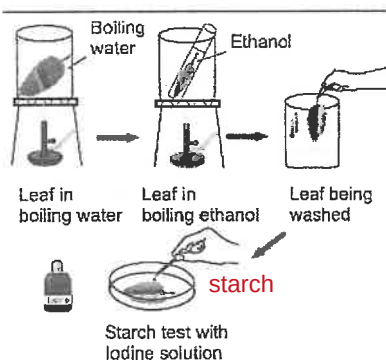


JH

Q: Draw and label one palisade plant cell – label the cell membrane, cell wall, nucleus, cytoplasm, vacuole and chloroplasts

photosynthesis
carbon dioxide
light
Light energy
chloroplasts
• Palisade cell drawing to label:
cell wall, cell membrane,
cytoplasm, nucleus, vacuole
and chloroplasts.
Iodine

- The **glucose** produced can be respired OR used to synthesise (make) st..... (as a food store), c..... (for cell walls), **protein and fats**. **cellulose**
- Orange i..... **solution** can be used to test a leaf for the presence of **starch** as it will change to **blue/black** (evidence that photosynthesis has taken place)



- Different foods contain different quantities of **starch** eg p..... contain a lot. Humans eat these foods a source of e.....

• Starch test diagram: boil the leaf in water, then in boiling ethanol (removes chlorophyll), wash it, then test with iodine solution - blue/black shows starch is present.

4. Factors affecting photosynthesis

- Pondweed is a water plant that released bubbles of **oxygen** when **photosynthesizing**
- The rate of photosynthesis can be measured by
.....
- L....., C..... d..... and an **optimum temperature** are all required for efficient bubble production **light intensity**

These **factors** can all affect the **rate of photosynthesis**.

5. Bio fuels

- A **fuel** is a source of chemical e.....
- Fuels are burnt (combusted) to produce heat
- **Biofuels** are produced from **readily replenished p**.....
- **Biofuels** still produce c d..... when burnt but are an alternative 'cleaner' source of fuel to **fossil fuels** because the plants used absorb c..... d..... when they grow.
- The main **biofuels** are e..... **The main biofuels are bioethanol (from sugar cane/maize) and biodiesel (from plant oils).**

Q: Can you think of any more advantages or disadvantages?

Advantages of Biofuels	Disadvantages of Biofuels
• Renewable energy sources • Biofuels are 'carbon neutral' as the carbon dioxide released on burning is removed by the growing plants • They are readily available – plants can be grown all over the world (but oil only found in some countries) oxygen The rate of photosynthesis can be measured by counting the number of oxygen bubbles released per minute (or by collecting and measuring the volume of gas). energy plants	• Large areas of forest have been cleared to make room to grow biofuels • Lots of species lose their habitat • The decay and burning of vegetation to clear land increases the carbon dioxide in the air • The plants need land to grow so can compete with food supply carbon dioxide carbon dioxide More advantages: renewable, cheap, carbon neutral. More disadvantages: forests cleared destroy habitats, CO₂ released when land is cleared, and crops compete with food supply.

8B 18b Ecology Summary Sheets

Ecological relationships

Habitats and environments

A **habitat** is the area where an organism lives. Non-living factors in a habitat are called **physical environmental factors**. Examples include how light it is and what the temperature is. These make up the surroundings, or **environment**, of an organism.

Small areas in a habitat are called **microhabitats**. For example, the fur on a fox is a microhabitat.



The fox's fur is a microhabitat.

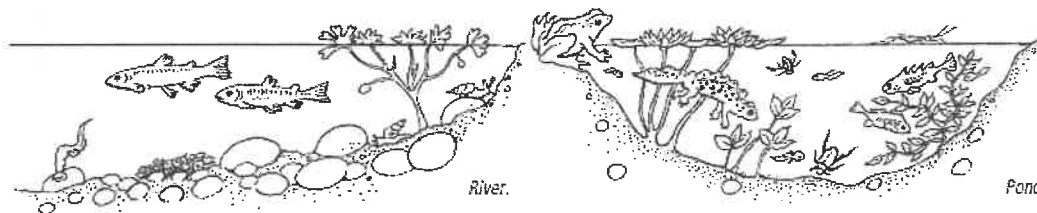
A woodland habitat.

All the plants and animals that live in a habitat make up a **community**. Within a community, the total number of one species is called a **population**. There will be a population of foxes in a wood.

In order to survive in a habitat, organisms need various **resources**. An animal needs food, water, oxygen, shelter and it needs to find a mate to reproduce. Plants need light, water and carbon dioxide in order to make food. They also need mineral salts (nutrients), oxygen and space to grow.

Adaptations

In any habitat, the organisms living there must be **adapted** to survive the environmental conditions within that habitat. Members of the same community may have similar **adaptations** to cope with the problems of their habitat. For example, many small animals and plants which live in fast-flowing rivers are adapted to stop them being swept away. In ponds, free-swimming animals and floating plants can survive because there is no current to wash them away.

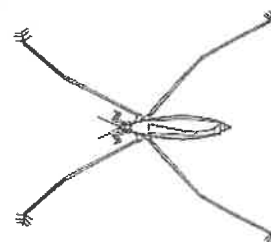


River.

Pond.

All the places in an environment where an organism is found is called its **distribution**. For example, in a pond habitat pond skaters are found on the surface of the water. They are adapted to living here because they have bristles on the ends of their legs, which prevent them from breaking the surface film of water. They feed on dead insects floating on the water.

Organisms that are better adapted to survive in an area will have a better chance of survival.



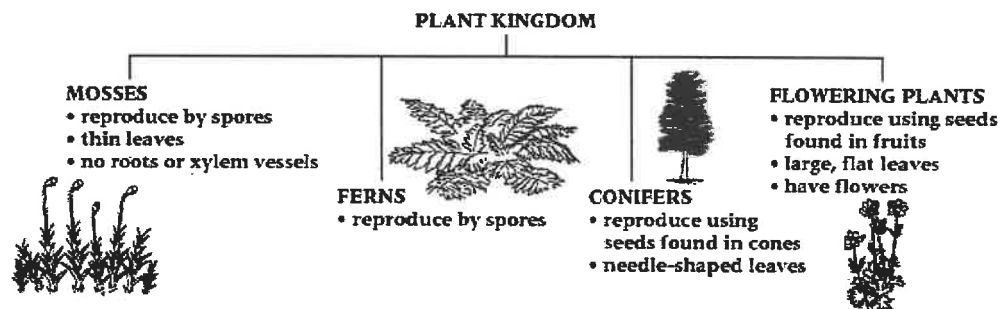
A pond skater.

Classifying living organisms

Each different type of organism is called a **species**. There are so many species that we need to put them into groups. This is called **classification**. The largest groups are called **kingdoms** and the biggest of these are the **plant kingdom** and the **animal kingdom**. The Summary Sheets for Unit 7D *Variation and classification* show how animals are classified into groups.

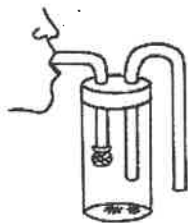
The main difference between plants and animals is that plants can make their own food by **photosynthesis**.

There are four main plant groups.

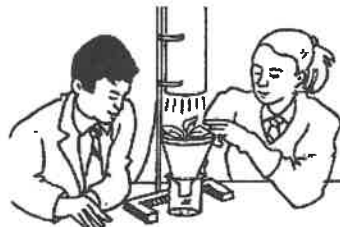


Sampling methods

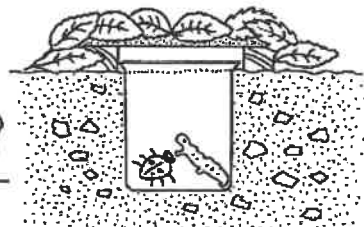
Ecologists are scientists who study habitats. They catch organisms using various **sampling techniques**, then use keys and field guides to identify them.



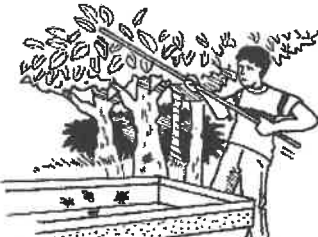
A potter is used to collect small invertebrates.



A Tullgren funnel is used to collect invertebrates from leaf litter.



A pitfall trap traps small animals that crawl along the ground.



Tree beating is used to collect animals that live in trees or bushes.



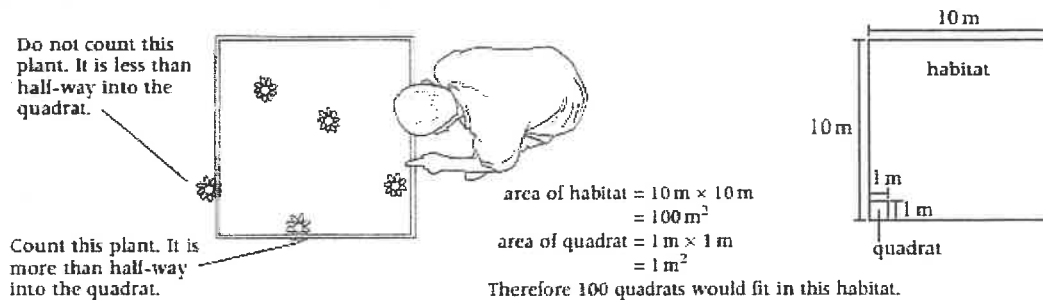
Pond dipping is used to collect organisms from different parts of a pond.



A sweepnet is used to collect organisms from long grass.

Ecologists often need to know the size of a population. It would be impossible to count all the organisms in a habitat, so they take **samples** and then **estimate** the total number of organisms.

A **quadrat** is a sampling square used to estimate plant populations. The quadrat is placed randomly on the ground in different parts of the habitat and the number of plants inside it is counted each time.



The more samples that are taken, the more reliable an estimate will be.

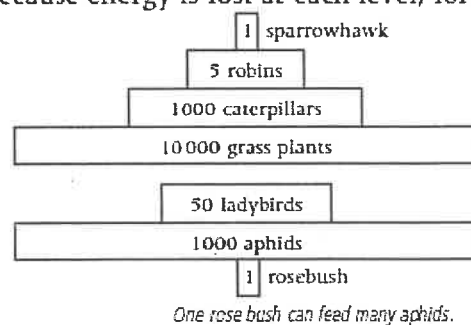
Ecologists also measure the physical environmental factors, like how warm it is and how much oxygen is dissolved in pond water, using sensors and dataloggers. They then look to see if there are any links between the factors, such as, the warmer the water the lower the oxygen level is.

Feeding relationships

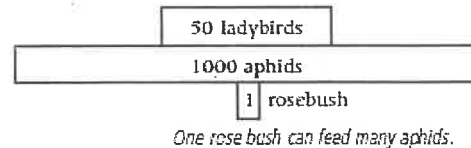
Food chains and **food webs** show the feeding relationships between different organisms in a habitat. (See Summary Sheets for Unit 7C *Environment and feeding relationships*).

The numbers of organisms at each level in a food chain can be shown as a **pyramid of numbers**. The size of the bars shows the number of organisms. Usually there are fewer organisms as you go along a food chain because energy is lost at each level, for example, for movement. Sometimes the pyramid has an unusual shape if the organisms are very different in size.

grass → caterpillar → robin → sparrowhawk



rose bush → aphids → ladybirds



Populations

The size of a population is affected by several factors.

- Animals compete with each other for resources such as food, water and shelter.
- Plants compete for light, water, nutrients (mineral salts) and space.
- If there are not enough resources the population will decrease.
- Disease can kill organisms.
- The populations of predators and prey are linked. When there are a lot of prey organisms, the number of predators increases because they have plenty of food. This decreases the number of prey, which then leads to a decrease in the number of predators.
- Harsh weather conditions can reduce populations.

Living organisms depend on others for their survival. For example, plants depend on insects for pollination.

18c

Photosynthesis

Water

Crops

Wheat

glucose

Light

Rice

Respiration

Biofuels

Plant

Biodiesel

anther

fertilization

Ethanol

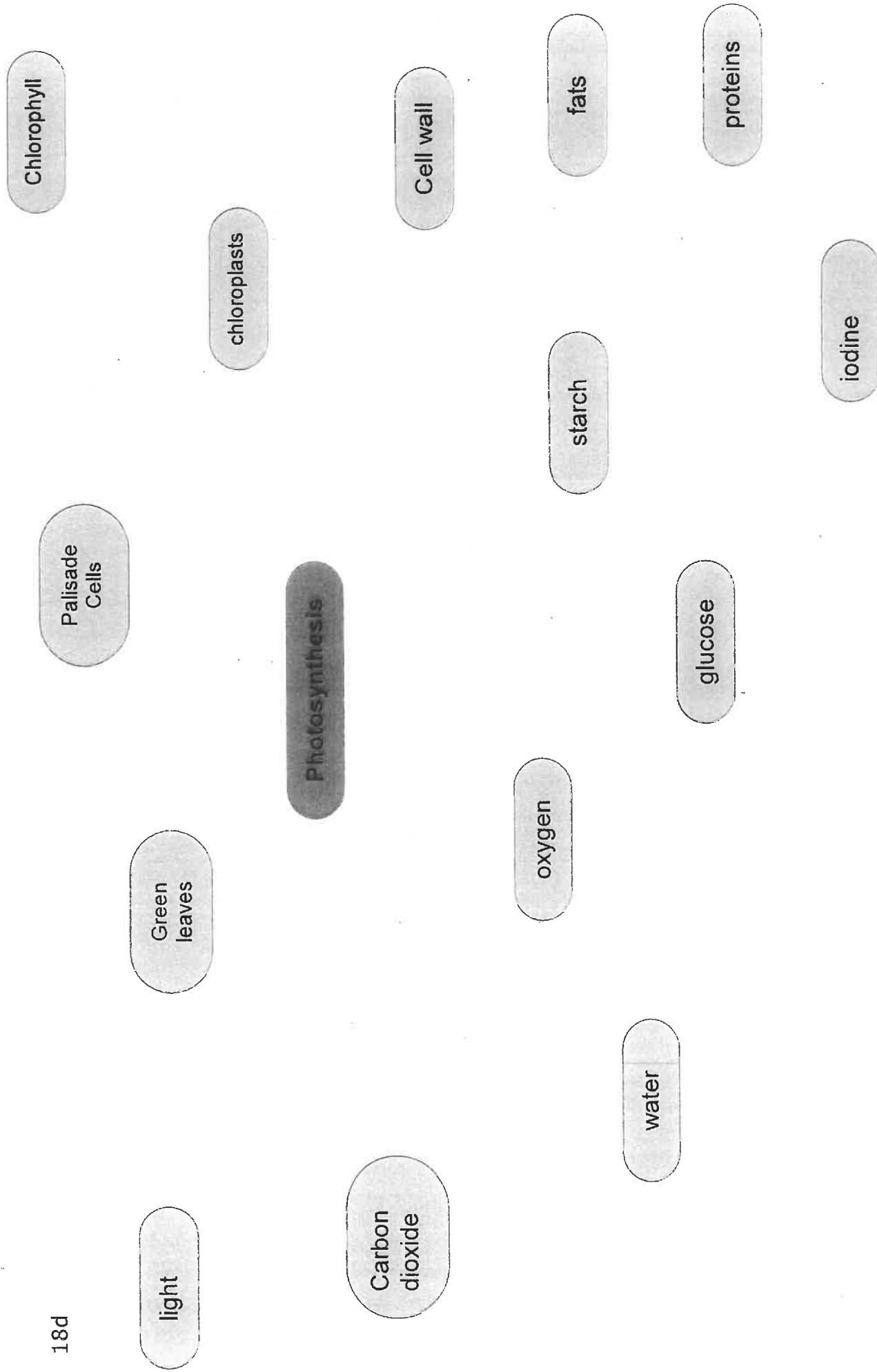
Pollen
grain

stigma

ovule

Pollen tube

embryo



18d

18e

Ecosystem

Community

Habitat

Environment

Transect

Pyramid of Numbers

Ecology

Prey

Population Size

Food Chain

Herbivore

Carnivore

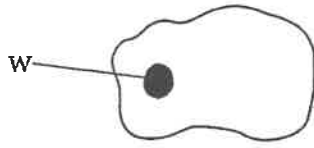
Predator

8B 18f Plants: Quick Quiz

On your answer sheet, write in or circle or highlight the correct letter for each question.

A) Revision of cells

- 1 On this drawing of an animal cell, what is part labelled 'W'?



- A cell surface membrane **C nucleus**
B chloroplast D heart

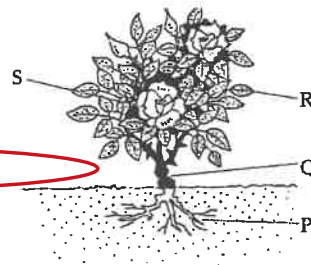
- 2 On the drawing of the animal cell, what does part 'W' do?

- A makes energy B makes food
C controls the cell D holds the cell together

B) Photosynthesis

- 1 Where does photosynthesis occur in a rose bush?

- A in all the parts
B in parts R and S only
C in part R only
D in part Q only



- 2 Plants often turn some of the food they make into another substance for storage. The name of this substance is:

- A glucose. **B starch.**
C fibre. D pasta.

- 3 Which of these shows the correct word equation for photosynthesis?

- A glucose + carbon dioxide → water + oxygen (+ energy)
B glucose (+ energy) → oxygen + glycerol
C water + carbon dioxide (+ energy) → glucose + oxygen
D chlorophyll + carbon dioxide → water + glycogen

- 4 Aerobic respiration is a process which:

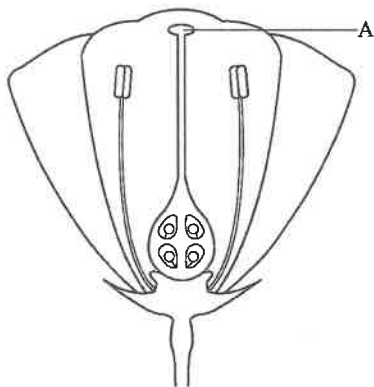
- A stores chemical energy in food.
B makes food for plants living under water.
C does not happen in plants but only in humans when they exercise hard.
D releases energy from food.

- 5** Leaves are often wide so that they can:
- A absorb as much sunlight as possible.
 - B catch as much rain as possible.
 - C provide shelter for animals living under them.
 - D float to the ground when they drop in autumn.
- 6** Gases get into and out of a leaf by:
- A diffusion.
 - B absorption.
 - C conduction.
 - D convection.
- 7** Look at this list of substances.
glucose fat cellulose protein
Which of these substances can plants make?
- A glucose and cellulose only
 - B glucose, cellulose and fat
 - C glucose, cellulose and protein
 - D all of them
- 8** Plants need cellulose to:
- A store energy.
 - B repair damaged cells.
 - C make cell walls.
 - D make cell surface membranes.
- 9** Choose from the following list the correct products of photosynthesis.
- A water and carbon dioxide
 - B glucose and carbon dioxide
 - C glucose and oxygen
 - D water and carbon dioxide
- 10** In plants starch is used:
- A to make flowers colourful.
 - B as a store of energy.
 - C to absorb water from the soil.
 - D to make leaves green.
- 11** Look at these chemical processes.
photosynthesis respiration burning fuels
Which of these processes add carbon dioxide to the air?
- A respiration and burning
 - B respiration only
 - C photosynthesis and respiration
 - D burning only

- 12 Plants are important in keeping the atmosphere in balance because:
- A they use up oxygen.
 - B they use up carbon dioxide.
 - C they provide food for animals and humans.
 - D they use water.
- 13 Which of these would **not** be a disadvantage of cutting down forests?
- A Logging companies could make more profits.
 - B There would be fewer trees to keep the atmosphere in balance.
 - C Some species might become extinct.
 - D Floods might wash away the soil.
- 14 Which of the following environmental conditions will **not** increase the rate of photosynthesis in a crop plant?
- A well-watered soil
 - B warm weather
 - C bright sunshine
 - D windy weather

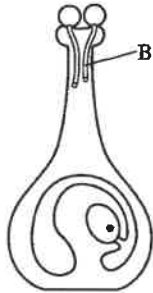
C) Fertilisation

- 1 On the diagram the part labelled with the letter 'A' is the:



- A style.
 - B ovary.
 - C stigma.
 - D anther.
- 2 Pollination is:
- A the carrying of pollen from an anther to a stigma.
 - B when two sex cells join together.
 - C when an insect eats pollen.
 - D when harmful substances enter the air.

- 3 Fertilisation is:
- A the carrying of pollen from an anther to a stigma.
 - B when two sex cells join together.
 - C when an insect eats pollen.
 - D when a seed starts to grow.
- 4 In the diagram, the part labelled 'B' is:



- A an ovary.
- B an anther.
- C a xylem tube.
- D a pollen tube.

D) Ecology

1. The five groups of vertebrates are:
- A chickens, dolphins, eagles, turtles.
 - B mammals, reptiles, penguins, worms, bacteria.
 - C amphibians, birds, reptiles, mammals, fish.
 - D fish, amphibians, mammals, snakes, insects.
2. All arthropods have:
- A four legs and two eyes.
 - B bodies divided into five sections.
 - C jointed legs.
 - D a 'fleshy pad'
3. Seeds are:
- A made by flowering plants and conifers.
 - B made by certain kinds of animal.
 - C only found in apples.
 - D made by excretion.
4. Plants can move water to their leaves in:
- A straw vessels.
 - B xylem vessels.
 - C dermatic vessels.
 - D leaf vessels.

5. Which three things are all physical environmental factors?

- A light intensity, temperature and oxygen concentration
- B number of predators, temperature and food supply
- C number of plants, number of predators and number of prey organisms
- D number of dead organisms, temperature and oxygen concentration

6. Which sampling technique would you use to find small animals in leaf litter or in a soil sample?

- A pond dipping
- B tree beating
- C a sweepnet
- D a Tullgren funnel

7. A garden measures 10m by 5m. A 1 m² sample is taken and found to contain two dandelion plants. What would you *estimate* the population of dandelion plants to be in the garden?

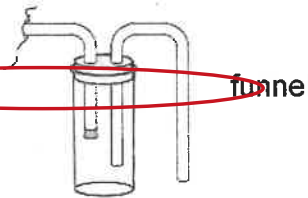
- A 2
- B 45
- C 50
- D 100

8. What is a person who studies habitats called?

- A a lumberjack
- B an ecologist
- C an echinoderm
- D a physicist

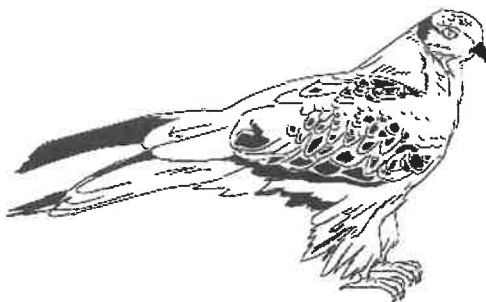
9. What is this piece of equipment called?

- A a pooter
- B a shooter
- C a sweepnet
- D a Tullgren funnel



10. Use the key to work out what bird this is.

- A European jay
- B Kite
- C puffin
- D wood duck



- | | | |
|---|---|--|
| 1 | <div style="display: inline-block; vertical-align: top;"> <div style="border-left: 1px solid black; padding-left: 5px;">Feet have claws.</div> <div style="border-left: 1px solid black; padding-left: 5px;">Feet are webbed.</div> </div> | <div style="display: inline-block; vertical-align: top;"> <div>go to 2</div> <div>go to 3</div> </div> |
| 2 | <div style="display: inline-block; vertical-align: top;"> <div style="border-left: 1px solid black; padding-left: 5px;">Has a hooked beak.</div> <div style="border-left: 1px solid black; padding-left: 5px;">Does not have a hooked beak.</div> </div> | <div style="display: inline-block; vertical-align: top;"> <div>Kite</div> <div>European Jay</div> </div> |
| 3 | <div style="display: inline-block; vertical-align: top;"> <div style="border-left: 1px solid black; padding-left: 5px;">Has stripes on its beak.</div> <div style="border-left: 1px solid black; padding-left: 5px;">Has no stripes on its beak.</div> </div> | <div style="display: inline-block; vertical-align: top;"> <div>Puffin</div> <div>Wood duck</div> </div> |

11. Some smaller plants that grow in woods have big leaves. This is because:

- ~~A there is not much light and bigger leaves can trap more light.~~
- B they need to provide shelter from the rain for small animals.
- C less rain gets through the trees and the big leaves help to absorb more rainwater.
- D they have got more room to spread out.

12. Here is a list of things found in a particular habitat.

blackbird rotting log snail
bracken plant rock tree

Which of these things form a community?

- A the rock and the rotting log
- B the rock, the log, the bird and the snail
- ~~C the bird, the snail, the bracken plant and the tree~~
- D all of them

13. What does the word 'distribution' mean?

- A Where organisms are found in a habitat.
- B The total number of organisms living in a habitat.
- C The different species that live in a habitat.
- D The different tribes that live in a habitat.

The different species (populations) living in the habitat - all the living organisms, e.g. blackbird, snail, bracken plant and tree (option D).

14. Which of these things do plants compete for?

- oxygen water heat light
- A all of them
 - B only oxygen
 - C oxygen and heat
 - ~~D water and light~~

light, water, space and mineral ions - plants compete for these, not for heat or oxygen.

15. Which of these things will **not** cause the number of rabbits in a population to decrease?

- A There are more lettuces for rabbits to eat.
- B More rabbits are eaten by foxes.
- C Less rabbits catch a fatal disease.
- D More rabbits catch a fatal disease.

16. Three resources that a species of rabbits needs to survive are:

- A food, oxygen and carbon dioxide.
- B food, water and a mate
- C water, light and carbon dioxide.
- D oxygen, carbon dioxide and shelter.

17. Which of the following is a producer?

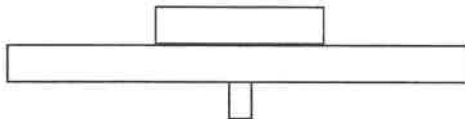
- A blue tit
- B cow
- C mushroom
- D rose bush

A green plant or alga (e.g. grass, lettuce) - a producer makes its own food by photosynthesis and starts the food chain.

18. In science, what is a 'consumer'?

- A Someone who watches Watchdog on television.
- B An animal that eats other organisms.
- C A plant that does not need to eat anything.
- D A plant that starts a food chain

19. This diagram shows the number of organisms in a food chain. Which of the following food chains does it represent?



- A grass → rabbit → fox
- B grass → sheep → tick
- C oak tree → caterpillar → bird
- D lettuce → slug → bird

20. The numbers of organisms in a food chain can be shown using:

- A a number chart
- B a triangle of numbers
- C a pyramid of numbers
- D a square of numbers

The food chain with the most producers at the base and fewest top predators, e.g. lettuce → slug → bird.

Pyramid of numbers - each bar shows the number of organisms at that level; numbers fall up the chain because energy is lost (respiration, movement, waste heat) at each stage.

Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p49: Yes

p49: Yes

p76: Microhabitat - a small area in a habitat where organisms live.

p76: a 150 lions.

p76: Extension: food (prey) supply, disease, hunting/poaching, drought, competition, predators, loss of habitat.

p86: When a seed germinates, the seedling emerges from the seed coat (it grows out of the seed).

p90: Revision summary page - complete. Key points: classification, kingdoms, four plant groups (mosses, ferns, conifers, flowering plants), and sampling methods (pitfall traps, Tullgren funnel, tree beating, pond dipping, sweep net).